PALEOMAGNETISM AND AGE DETERMINATIONS OF THE DECCAN TRAPS (INDIA): RESULTS OF A NAGPUR-BOMBAY TRAVERSE AND REVIEW OF EARLIER WORK

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Abstract. Review of available radiometric age determinations of the Deccan traps (India) shows a spectrum of K-Ar ages that is highly polluted by argon loss. Stepwise ^{40}Ar–^{39}Ar age determinations include estimates of data quality and thus avoid contaminated results. The absolute age of the Deccan traps determined using 22 ^{40}Ar–^{39}Ar plateau age spectra is 65.5 ± 2.5 Ma. Paleontological data on infratrappean and intertrappean sediments constrain Deccan age to between the *A. mayaroensis* zone, in the Upper Maestrichtian (about 67 Ma), and the P2 foraminifer zone, in the Lower Paleocene (about 60.5 Ma). Paleomagnetic study of a Nagpur-Bombay traverse (preliminary results of which were used by Courtillot et al. (1986*a*, *b*) for a general discussion about Deccan volcanism and the Cretaceous-Tertiary boundary) is presented in detail. All available paleomagnetic results from the Deccan traps (563 flows) are then compiled. Results considered to be transitional or to come from suspicious sites are removed leaving 485 flow results. This extensive data set from a single geological unit allowed us to look in some detail at its statistical distribution. The virtual geomagnetic poles (VGP) are approximately Fisher distributed but present a complex asymmetry. No regional variation can be seen (to within paleomagnetic uncertainties). Although the 3.5° angular difference between the separate normal (pole) and reversed (antipole) data is not statistically significant, it can be explained by either a 2.1 m.y. drift along the apparent polar wander path (APWP) of the Indian plate assuming a normal-reverse-normal (*N-R-N*) magnetostratigraphy, or a 3.5% contamination by a present field overprint, or a slight nondipole field component. A quality coefficient has been assigned to each result on the basis of existence and value of published 95% confidence angle. Because the normal and reversed mean poles become more precisely antipodal with higher-quality data and with more recent publication date (as a consequence of the evolution of paleomagnetic techniques), we favor the overprint hypothesis. The angular standard deviation of our VGP set is 20% larger than the value predicted by the paleosecular variation model of McFadden and McElhinny (1984). Finally, our best estimate of the overall mean pole is located at 281.3°E, 36.9°N, A_{95} = 2.4° (calculated from the 163 highest-quality flow VGPs). It is in remarkable agreement with the reference APWP of Besse and Courtillot (1991), leading to an independent estimate of the age of Deccan volcanism at 67.2 ± 6.6 Ma. The same comparison following Acton and Gordon (1989) provides an estimated age at 67 ± 5 Ma. These estimates which are based on paleomagnetic data only are in perfect agreement with both radiometric and paleontological ages.

1. INTRODUCTION

The Deccan traps of central India and southernmost Pakistan comprise a thick succession of subaerially erupted basaltic flows which cover an area of about half a million square kilometers and may originally have covered over 1.5×10^6 km^2 [*Krishnan*, 1960; *Raja Rao et al.*, 1978], for a total volume possibly in excess of 2×10^6 km^3 [*Pascoe*, 1964]. New paleomagnetic, paleontological, and radiochronological data indicate that the Deccan traps may have erupted in a rather short interval of less than 1 m.y., straddling the Cretaceous-Tertiary boundary (KTB) [*Courtillot et al.*, 1986*a*, *b*]. It has been argued that the flood basalts marked the first eruption on the Earth's surface of material related to the Réunion hotspot, indirectly resulting in the many biological and geological anomalies observed at the KTB, and followed shortly afterward by the rifting apart of the Arabian Sea [*Courtillot et al.*, 1986*a*, *b*; *Courtillot and Cisowski*, 1987; *Courtillot and Besse*, 1987]. The aim of this review is to present an up-to-date compilation of available age constraints and paleomagnetic data from a formation that may have resulted from the most significant volcanic event on Earth since the Triassic. Fieldwork by our group in India, between 1984 and 1986, was primarily aimed at sampling an extensive traverse from Nagpur to Bombay (Figure 1) in order to gather age data and paleomagnetic results that could be used both to complement and allow the critical evaluation of preexisting data.

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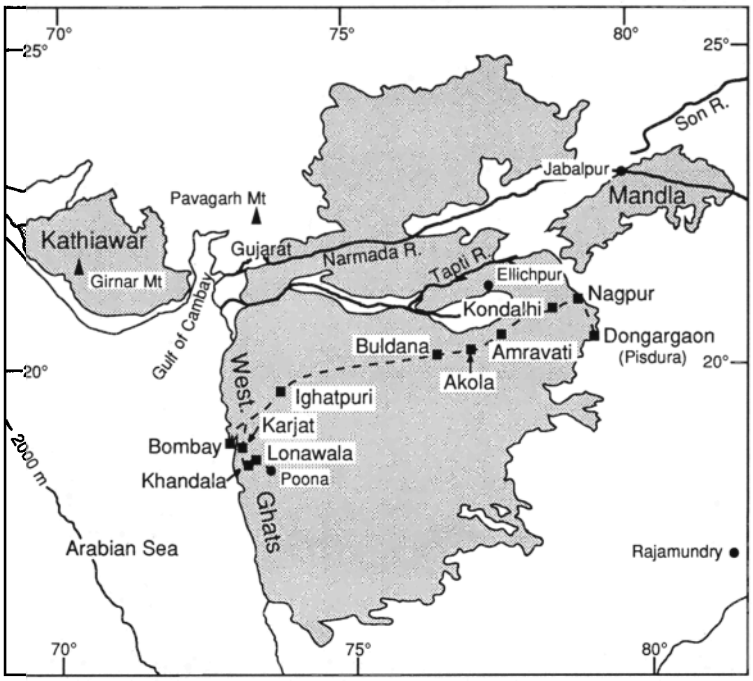


Figure 1. Geographical extent of present day outcrops of the Deccan traps with place names used in the paper. The paleomagnetic traverse for which new data are reported is indicated by a dashed line, and sampling sites are marked as full squares.

We first provide a brief account of relevant geological features of the Deccan traps. Then, we present a set of new K-Ar age determinations and a review of all other radiometric ages (^{40}Ar–^{39}Ar as well as K-Ar) available to us. We also briefly recall paleontological constraints [*Jaeger et al.*, 1989]. We next give a detailed report of our paleomagnetic measurements, which extends and complements the preliminary report of *Courtillot et al.* [1986*a*, *b*]. Then, we review the extensive set of published paleomagnetic results from the Deccan, incorporate our own results, and analyze the resulting data set, one of the most extensive for a single formation. We discuss the statistical distribution of virtual geomagnetic poles (VGPs) as a function of geographic location, magnetic polarity, quality of the original individual data sets and propose an overall mean pole for India at KTB time. This pole is compared to the synthetic apparent polar wander path (APWP) recently produced by *Besse and Courtillot* [1990] and transferred to India. Slight differences between mean poles for normal and reversed flows are discussed in terms of overprinting and/or continental drift. The Deccan pole is also transferred to the North American plate for comparison with the results recently obtained by *Acton and Gordon* [1989]. The angular standard deviation of VGPs is finally compared to the paleosecular variation (PSV) model of *McFadden and McElhinny* [1984].

2. GEOLOGY OF THE DECCAN TRAPS

Geological data and various interpretations of the Deccan traps have recently been reviewed by *Mahoney* [1988]. The immense lava pile is essentially flat lying, with dips of the order of 1° or less, except along the western coast and north into the Cambay graben and in the east-west (EW) trending Narmada valley (Figure 1). The lava sequence is thinnest in the east and thickens westward to more than 2 km in the Western Ghats. The maximum exposed thickness is 1700 m in the Ighatpuri area. The thickness appears to be controlled by the pre-Deccan topography and by faults confined to the areas mentioned above where significant dips are observed. In the east, many flows can be traced over distances well in excess of 100 km with no signs of dying out, implying surfaces of several 10^3 km^2 and possibly much more. Flows seem to be more numerous and to have more limited extent in the western Deccan. Systematic analysis of regional dip patterns is difficult and, when attempted, shows very large subdued, elongate domal structures [*Raja Rao et al.*, 1978]. Geochemical [*Beane et al.*, 1986; *Devey and Lightfoot*, 1986] and magnetostratigraphic [*Courtillot et al.*, 1986*b*] analyses appear to be the most promising to unravel large-scale structures (D. Vandamme et al., manuscript in preparation, 1991) and discriminate between a constructional or a subsequent tectonic origin of the dips. Although Deccan volcanism was traditionally interpreted to be of fissural type, the only identified focus of emission is the elongate dome found in the Western Ghats by *Raja Rao et al.* [1978] and *Beane et al.* [1986]. Older reports suggested a younging of the flows from east to west, but this notion is now largely rejected. It is not confirmed by radiometric or paleomagnetic analyses. Distinct sources, possibly including the Narmada "lineament," are therefore advocated.

The Deccan lavas are predominantly basalts with a great diversity in Nd, Sr, and Pb isotopic ratios. They are evolved lavas, enriched in Fe and Ti, and have undergone highly variable, sometimes considerable contamination by

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the continental crust or lithospheric mantle. A very large source region in a mantle with Nd characteristics intermediate between the present-day Central Indian ridge and Réunion hotspot is favored by *Mahoney* [1988]. The lava texture ranges from aphiric to doleritic. Phenocrists are mainly plagioclase and to a lesser extent olivine and clinopyroxenes. Their normative composition is predominantly tholeiitic in the east and southeast but varies from tholeiitic to alkaline in the west and northwest (Western Ghats, Kathiawar, Gujarat; see Figure 1). Differentiated rocks are rare and localized. Acidic lavas are confined to two areas, one from Bombay to Mount Pavagarh in the north, the other in the western part of the Narmada valley and in Kathiawar (Figure 1). The location of acidic rocks corresponds to that of positive gravity anomalies, which are interpreted as thick intrusions of mafic material or magma chambers,and may possibly be associated with the main eruptive sources of the Deccan traps.

3. AGE DETERMINATIONS

Alteration is an ubiquitous and annoying feature of the Deccan basalts. Extensive alteration is often obvious macroscopically, but even samples that may appear to be fresh in the field reveal alteration minerals, such as calcite and zeolites, and significant proportions of water. This can of course be a major difficulty for both paleomagnetic and geochemical analyses; age determinations are further impaired by the fine grained texture of the basalts and the virtual absence of argon-retentive phases. As a result, published K-Ar ages show a bewildering spectrum [*Courtillot et al.*, 1986*b*; *Baksi*, 1987; *Mahoney*, 1988]. In keeping with several previous authors, *Mahoney* [1988] cites a period of maximum outpouring of flood lavas from 66 to 57 Ma.

3.1. Radiometric Dating

Our own samples come from an EW traverse from Nagpur to Bombay (Figure 1), perpendicular to the tentative structures which had previously been suggested in most of the Deccan, except in the north near the Narmada valley. A total of 14 flows were sampled; it was very difficult to determine whether a given flow was encountered at more than one locality. Samples are from Dongargaon (two flows), Nagpur (two flows), Kondalhi (one flow), Amravati (one flow), Buldana (five flows), Ighatpuri (one flow) and Bombay (two flows) (see Figure 1). The K-Ar investigation was carried out on mineral fractions of plagioclase. The extracted minerals were usually obtained from the 100–160 μm sieve fraction by use of heavy liquids and magnetic separation. The analytical procedure, more extensively described elsewhere [*Westphal et al.*, 1979], is as follows. Potassium was measured by flame photometry with a lithium internal standard. Argon was extracted in a heat-resistant glass vacuum apparatus and determined by isotope dilution (^{38}Ar being used as a tracer) using a MS10S mass spectrometer. All samples were measured using the static method. The set of constants recommended by *Steiger and Jäger* [1977] was used for age calculation. Quoted uncertainties represent estimates of analytical precision at one standard deviation. They were calculated using the procedure given by *Cox and Dalrymple* [1967]. Results for 14 specimens were reported by *Courtillot et al.* [1986*b*] and an additional 15 are reported here (Table 1). Only one sample (DK16) showed macroscopic signs of alteration; in seven others (marked by an asterisk in Table 1) alteration was detected on thin sections, and the carbonate content left no doubt as to the importance of this alteration. Sample fractions were separated according to their magnetic susceptibility, and the magnetic fraction (marked by the letter "m" in Table 1) often yielded, for a given flow, the oldest age and the smallest uncertainty. Sample DK12 has an older nonmagnetic fraction, but the error bar is considerable (24 Ma), and the age (97 Ma) is well outside the range of all other samples. Microscopic observation revealed no significant evidence of alteration in this sample. Sample DK 17 also has an older age for the nonmagnetic fraction. Excluding values where alteration was found, and retaining only determinations made on magnetic fractions (quality *A* in Table 1), we end up with a total of 14 determinations for 11 flows, with an age range from 53.9 to 66.8 Ma, and an average 1σ uncertainty of 3 Ma (the minimum being 2.0 Ma). Our data are shown in Figure 2. We have integrated data with a Gaussian distribution to emphasize the statistical significance of age uncertainties. Each datum is given unit weight, which flattens the large uncertainty data and emphasizes the most precise results. The upper curve is for all determinations and the lower one for nonaltered magnetic fractions (shading). The lower curve, corresponding to the "best" data extends over 30 Ma, rising sharply between 70 and 63 Ma with a maximum around 62 Ma.

We compiled similar histograms from 70 previously published determinations of K-Ar ages [*Rama*, 1968; *Wellman and McElhinny*, 1970; *Kaneoka and Haramura*, 1973; *Agrawal and Rama*, 1976; *Wensink et al.*, 1977; *Alexander*, 1981; *Balasubrahmanyan and Snelling*, 1981] added to the 29 presented above. Most of these ages (specified in Table 2) have had to be corrected for the newer decay constants of *Steiger and Jäger* [1977]. Data selection and evaluation are not easy tasks, in part because of a lack of sufficient geochemical information in the original papers; we have only considered as less reliable the data identified as such by original authors (*B* quality in Table 2), for reasons related to sampling or experimental observations and briefly explained in the comment column of Table 2. *Balasubrahmanyan and Snelling* [1981] published ages obtained on pyroxene but used them as indications of Ar motion in the rocks: these ages are listed in Table 2 with *C* quality. In the case of total lack of author information about sample alteration we have considered the datings as being of *A* quality (comment "no indic." in Table 2).

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TABLE 1. K-Ar Age Determinations

Locality	*Site*	*Loc.*	*HF*	*Frac.*	*Mineral*	*K*	*Rad^{40}Ar*	*%rad^{40}Ar*	*Age, Ma*	*Quality*
Dongargaon	01	1	*tr*	*m*	*Pl*	0.257	3.008	66.0	66.0 ± 4.0	*A*
	01	1	*	*m*	*Pl*	0.706	7.726	72.8	62.0 ± 2.0	*B*
	02	2	*tr*	*nm*	*Pl*	0.139	1.044	14.6	42.6 ± 7.5	*B*
	02	2	...	*m*	*Pl*	0.327	3.412	53.2	59.2 ± 3.0	*A*
Nagpur	08	1	*tr**	*nm*	*Pl*	0.115	0.969	6.9	48.2 ± 9.0	*B*
	08	1	*tr*	*m*	*Pl*	0.139	1.518	37.7	61.7 ± 3.5	*A*
	08	1	...	*m*	*Pl*	0.291	3.009	46.9	58.8 ± 2.5	*A*
	04	2	*	*nm*	*Pl*	0.098	0.6862	6.5	39.7 ± 11.0	*B*
	04	2	*tr**	*nm*	*Pl*	0.106	0.7885	12.0	42.3 ± 7.0	*B*
	04	2	*tr*	*m*	*Pl + Ox*	0.133	1.441	44.9	61.5 ± 4.0	*A*
	04	2	...	*m*	*Pl*	0.365	4.310	54.0	66.8 ± 2.5	*A*
Kondalhi	12	?	*tr*	*nm*	*Pl*	0.049	0.844	6.4	97 ± 24	*B*
	12	?	...	*m*	*Pl + Ox*	0.198	1.881	40.5	53.9 ± 2.5	*A*
Amravati	13	?	*tr**	*nm*	*Pl*	0.074	0.2570	2.1	19.9 ± 7.5	*B*
	13	?	*tr*	*nm*	*Pl*	0.116	1.306	42.1	63.7 ± 4.5	*B*
Buldana	15	4	*tr*	*nm*	*Pl*	0.125	1.300	33.7	59.2 ± 4.5	*B*
	15	4	*tr**	*m*	*Pl*	0.190	2.028	53.1	60.5 ± 3.0	*B*
	...	5	...	*m*	*Pl*	0.242	2.314	44.8	54.2 ± 3.5	*A*
	...	6	*tr**	*nm*	*Pl*	0.308	2.972	20.3	54.8 ± 3.5	*B*
	...	6	*tr*	*m*	*Pl*	0.438	4.928	74.4	63.7 ± 2.0	*A*
	...	6	*tr*	*m*	*Pl*	0.340	3.965	71.8	66.1 ± 2.0	*A*
	16	7	*	*m*	*Pl*	0.280	2.539	43.3	51.6 ± 2.5	*B*
	17	9	*tr*	*nm*	*Pl*	0.116	1.313	31.3	64.0 ± 5.0	*B*
	17	9	*tr*	*m*	*Pl*	0.199	2.082	53.8	59.3 ± 3.0	*A*
Ighatpuri	...	?	*tr*	*nm*	*Pl*	0.183	2.296	32.8	71.1 ± 4.5	*B*
	...	?	*tr*	*m*	*Pl*	0.183	2.000	46.7	62.1 ± 3.0	*A*
Bombay	...	?	*tr*	*m*	*Pl*	0.189	2.649	33.4	62.8 ± 3.0	*A*
	...	??	*tr*	*m*	*Pl*	0.265	2.813	36.7	60.2 ± 2.5	*A*
	...	??	*tr*	*m + nm*	*Pl*	0.401	4.976	73.5	70.2 ± 2.5	*B*

Data are from this study and *Courtillot et al.* [1986*b*]; $\lambda_\varepsilon = 0.581 \times 10^{-10}$ a^{-1}, $\lambda_\beta = 4.962 \times 10^{-10}$ a^{-1}, ^{40}K/total $K = 1.167 \times 10^{-4}$. Site: number of the paleomagnetic site sampled in the same flow (see Table 4 and Figure 1 for location). Center dots indicate no paleomagnetic sampling in the site. Loc: local stratigraphic position of the studied flow (numbers increasing upward). A single question mark and a double question mark differentiate sampling sites without stratigraphic correlation. HF: treatment with hydrofluoridric acid; *tr* is the treated samples; an asterisk is the occurrence of calcite, eliminated by HF treatment, and/or significant amounts of altered plagioclases. Frac: studied fraction of samples (after separation); *m* is magnetic; *nm* is nonmagnetic. Mineral: studied mineral. *Pl* is plagioclase; *Ox* is oxide. *K* is given in weight percent; rad^{40}Ar is given in 10^{-11} moles/g; Age: uncertainty is given at the 1σ (68%) confidence level. Quality of age determination; *A*, no observed alteration; *B*, alteration observed on bulk samples or on thin sections.

We separately show in Figure 3 age histograms for basalts, alkaline, and differentiated rocks and dykes: the latter two categories might have been expected to provide younger ages than the former. In fact, the latter two show a first peak which corresponds to the main peak of the basalt spectrum and a second weaker one corresponding to the young tail of the basalt spectrum. The small number of samples in the dyke category does not allow one to determine whether the slightly younger ages in the older peak are statistically significant. Consequently, it seems reasonable to study the three rock types together. The corresponding histogram (Figure 4) is similar to that of Figure 2. The sharp onset at 70 Ma is apparent, the peak position is almost identical around 62.5 Ma, and the tail is somewhat longer, extending to 30 Ma. This is clearly due to data that have suffered argon loss and yet were graded as being of good quality by their authors. Almost everyone now agrees that ages younger than 50–55 Ma probably correspond to some amount of argon loss [e.g., *Courtillot et al.*, 1986*b*; *Baksi*, 1987]. *Baksi* [1987] discusses some of the available data in terms of thin section examinations and atmospheric argon content and finds that "reliable" results are confined to the interval from 60 to 67 Ma. This compares well with the width of the histogram peak of Figure 4 at half amplitude (57.5–68.0 Ma) but fails to underline the sharp rise and significant asymmetry of this histogram [*Courtillot et al.*, 1986*b*].

The technique of ^{40}Ar–^{39}Ar plateau ages is more efficient to determine the alteration level. A plateau is that part of an age-spectrum diagram composed of continuous gas fractions that together represent more than 50% of the total ^{39}Ar released from the sample and for which no difference in age can be detected between any two fractions at the 95% level of confidence [*Fleck et al.*, 1977]. Up to a few years ago, very few studies of this kind were available. Moreover, *Baksi* [1987, 1988] criticized

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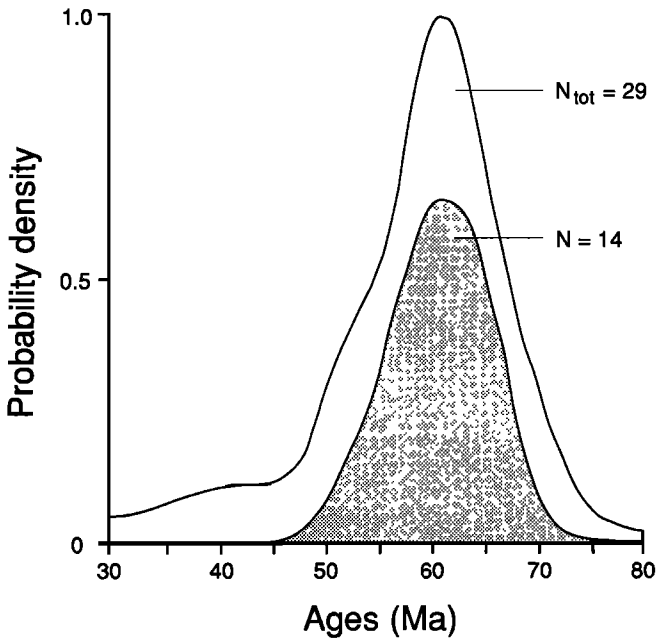


Figure 2. Histogram of K-Ar ages obtained along the Dongargaon-Bombay traverse reported by *Courtillot et al.* [1986*a*, *b*] and in this paper. Each datum is given unit weight and represented by a Gaussian distribution with standard deviation equal to the age uncertainty (1σ). The shaded area corresponds to determinations made from non magnetic fractions in which no evidence for alteration was found (A quality data in Table 1). N_{tot}: total number of measurements (A and B qualities); N: number of A quality measurements.

the few ^{40}Ar–^{39}Ar ages published by that time. Baksi rejected one of three plateau ages published by *Kaneoka* [1980], arguing for an unjudicious choice of his monitor sample (Bern 4M muscovite), which artifically created a plateau. Indeed, Kaneoka, in the error calculations, included a too large value for σ*j* (uncertainty in determination of the irradiation parameter). Usually, following *Berger and York* [1970], most authors assume σ*j* = 0. Also, Baksi recalculated the two other determinations, correcting σ*j*. *Baksi* [1987] recalled that *Venkatesan et al.* [1986] chose the same monitor sample and he showed that the initial ^{40}Ar–^{39}Ar ratios of all their samples indicated argon loss. As a result, only two plateau ages determined by *Kaneoka* [1980] and corrected by *Baksi* [1987, 1988] at 68.6 ± 2.4 Ma and 63.2 ± 2.3 Ma could be retained. Subsequently, several laboratories organized ^{40}Ar–^{39}Ar dating campaigns, and 28 plateau ages have been obtained [*Duncan and Pyle*, 1988; *Courtillot et al.*, 1988; *Pande et al.*, 1988; M. J. Kunk, personal communication, 1988; *Baksi*, 1989; *Shaw et al.*, 1990]. They are listed in Table 3. Although the measurements of *Pande et al.* [1988] come from the same laboratory as those of *Venkatesan et al.* [1986], the standard specimen was changed (to the 520.3 ± 1.8 Ma old Minnesota hornblende) and, moreover, ^{40}Ar–^{39}Ar initial ratios look correct, at least for the three samples (KB85-58, BHUJF, and KB83-13) considered as clean by the authors.

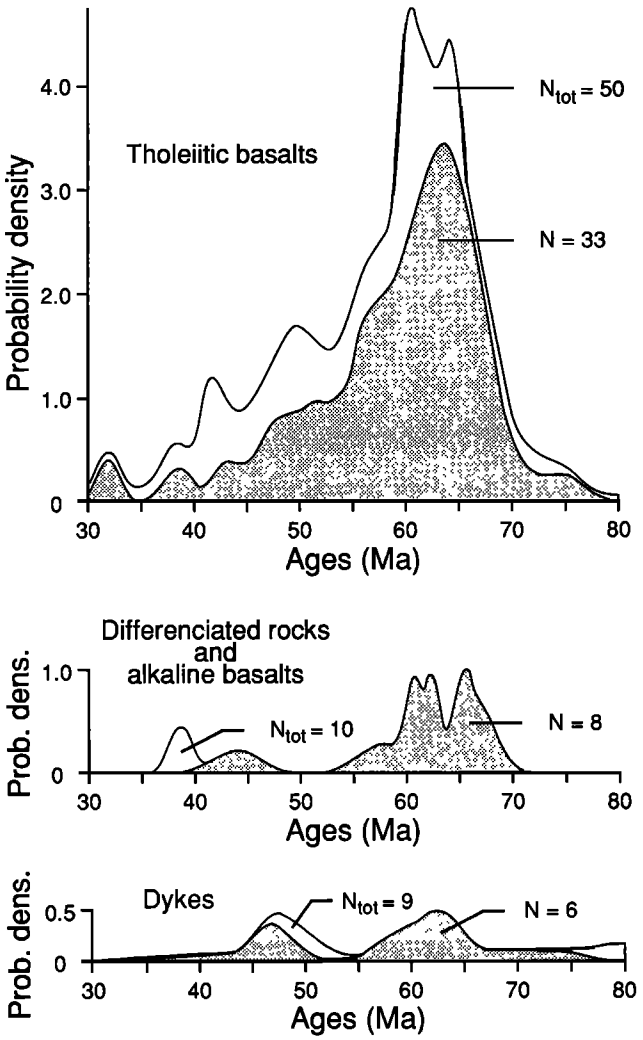


Figure 3. Same as Figure 2 with all K-Ar ages reviewed in this paper and listed in Table 2. The shaded area corresponds to the best data (A quality in Table 2). The three diagrams correspond to distinct sample petrologies: tholeiitic basalts, differentiated rocks and alkaline basalts, and dykes. N_{tot}: total number of measurements (A and B qualities); N: number of A quality measurements.

We used the same criteria to make a selection among the complete data set: results with ^{40}Ar–^{39}Ar ratio smaller, or much larger than 296, and/or those criticized by their authors themselves are considered as less reliable (*B* quality in Table 3) and are not used for mean age determination. We only used plateau ages in the following discussion and in Figure 5; however, the published measurements which did not yield plateau determinations are listed in Table 3 (*C* quality). In the case of ^{40}Ar–^{39}Ar datings the number of measurements on alkaline and differentiated rocks or on dykes is insufficient to allow a separate study of the various rock types. The histogram of all measurements (Figure 5; 22 are of *A* quality) is roughly symmetrical, with multiple peaks, and covers the main part of the sharp rise of the K-Ar spectrum (between 63 and 68 Ma at half amplitude); this is consistent with a systematic

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TABLE 2. Summary of Previous K-Ar Age Determinations

Locality	*Rock*	*Mineral*	*K*	*rad^{40}Ar*	*%rad^{40}Ar*	*Age, Ma*	*Quality*	*Comments*
			Rama [1964] (Corrected)					
Malad	*tr*	W.R.				61.5 ± 3.0	*A*	no indic.
Andheri	*ba*	W.R.				46.2 ± 3.0	*A*	no indic.
Pavagarh	*ba*	W.R.				66.6 ± 4.0	*A*	no indic.
Pavagarh	*rh*	W.R.				44.1 ± 2.0	*A*	no indic.
Lonawala	*dy*	W.R.				43.1 ± 6.0	*A*	no indic.
			Wellman and McElhinny [1970] (Corrected)					
Pavagarh	*rh*	W.R.	5.121	5.663	92.0	62.7 ± 0.6	*A*	
Pavagarh	*rh*	W.R.		5.511	95.2	61.0 ± 0.6	*A*	
Pavagarh	*ba*	W.R.	0.789	0.861	67.5	61.8 ± 0.7	*B*	alteration
Pavagarh	*ba*	W.R.	0.654	0.745	64.9	64.8 ± 0.6	*B*	alteration
Gulbarga	*ba*	W.R.	0.239	0.254	85.4	60.6 ± 0.6	*B*	alteration
Gulbarga	*ba*	W.R.	0.234	0.259	54.1	60.5 ± 0.6	*B*	alteration
Girnar	*di*	W.R.	2.334	2.753	76.1	65.7 ± 0.7	*A*	
			Kaneoka and Haramura [1973] (Corrected)					
Mahabaleshwar	*ba*	W.R.	0.216	1.577	24.8	41.6 ± 0.8	*B*	high $H_2O(+)$
Mahabaleshwar	*ba*	W.R.		1.629	83.8	43.1 ± 2.6	*B*	high $H_2O(+)$
Mahabaleshwar	*ba*	W.R.	0.488	4.854	30.6	56.5 ± 1.2	*A*	
Mahabaleshwar	*ba*	W.R.	0.340	3.423	78.9	57.1 ± 2.7	*A*	
Mahabaleshwar	*ba*	W.R.	0.249	2.170	32.3	49.6 ± 1.1	*B*	alteration
Mahabaleshwar	*ba*	W.R.	0.133	1.115	91.9	48.0 ± 5.6	*B*	alteration
Mahabaleshwar	*ba*	W.R.	0.257	2.827	83.3	62.3 ± 3.7	*A*	
Mahabaleshwar	*ba*	W.R.	0.423	4.703	44.7	63.1 ± 1.4	*A*	
Mahabaleshwar	*ba*	W.R.	0.166	1.319	52.4	44.7 ± 3.1	*B*	alteration
Amboli	*ba*	W.R.	0.374	3.998	70.2	61.7 ± 2.1	*B*	alteration
Amboli	*ba*	W.R.	0.257	2.341	88.8	52.5 ± 4.8	*B*	high $H_2O(+)$
Bombay	*mu*	W.R.	1.20	8.136	47.2	38.7 ± 0.9	*B*	alteration
Girnar	*di*	W.R.	2.56	30.23	16.7	67.0 ± 1.3	*A*	
Pavagarh	*ba*	W.R.	0.589	6.998	57.0	67.3 ± 1.7	*A*	
Pavagarh	*an*	W.R.	0.598	7.161	31.0	67.9 ± 1.4	*A*	
			Wensink et al. [1977] (Corrected)					
Mahabaleshwar	*ba*	W.R.	0.252	2.70	39			
			0.253	2.93	33			
				2.80	86	63.7 ± 3.1	*A*	no indic.
Mahabaleshwar	*ba*	W.R.	0.313	3.30	36			
			0.312	3.43	38	61.4 ± 3.0	*A*	no indic.
Kambatgi	*ba*	W.R.	0.204	2.363	46			
			0.205	2.435	42	67.0 ± 3.3	*A*	no indic.
Kambatgi	*ba*	W.R.	0.160	1.625	59			
			0.160	1.683	50	59.2 ± 2.9	*A*	no indic.
Panhala	*ba*	W.R.	0.133	1.360	53			
			0.134	1.320	48	57.4 ± 2.8	*A*	no indic.
Amba	*ba*	W.R.	0.568	6.55	24			
			0.568	6.50	23	65.6 ± 2.0	*A*	no indic.
Amba	*ba*	W.R.	0.367	4.30	31			
			0.369	4.20	38	65.9 ± 2.0	*A*	no indic.
Amba	*ba*	W.R.	0.154	1.350	73			
			0.156	1.308	73	49.2 ± 2.4	*A*	no indic.
Phonda	*ba*	W.R.	0.261	3.13	48			
			0.261	3.35	40			
				2.93	22	64.6 ± 5.0	*A*	no indic.
Ellichpur	*ba*	W.R.	0.138	1.470	55			
			0.138	1.495	48	61.4 ± 1.8	*A*	no indic.
Pohor	*ba*	W.R.	0.280	3.03	28			
			0.280	3.00	25	61.1 ± 1.8	*A*	no indic.
Nagpur	*ba*	W.R.	0.223	2.323	56			
			0.224	2.263	40	58.8± 2.9	*A*	no indic.
Nagpur	*ba*	W.R.	0.275	2.78	53			
			0.276	2.90	57			
				2.55	48	57.1 ± 4.5	*A*	no indic.

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TABLE 2. Continued

Locality	*Rock*	*Mineral*	*K*	*rad*^{40}*Ar*	*%rad*^{40}*Ar*	*Age, Ma*	*Quality*	*Comments*
		Agarwal and Rama [1976] Excerpted From *Alexander* [1981] (Corrected)						
Bhor Ghat	*ba*	W.R.				64.2 ± 1.5	*A*	no indic.
Satara	*ba*	W.R.				64.2 ± 1.7	*A*	no indic.
Chindwara	*ba*	W.R.				48.2 ± 1.5	*A*	no indic.
		Alexander [1981] (Corrected)						
Sagar	*ba*	W.R.	0.528	5.159	65.1	55.5 ± 1.8	*A*	
Sagar	*ba*	W.R.	0.182	1.370	47.0	42.8 ± 1.3	*A*	
Sagar	*ba*	W.R.	0.211	1.736	38.3	46.9 ± 1.5	*A*	
Sagar	*ba*	W.R.	0.289	2.602	44.9	51.3 ± 1.6	*A*	
Sagar	*ba*	W.R.	0.545	4.909	54.7	51.3 ± 1.6	*A*	
Koyna	*ba*	W.R.	0.200	1.120	42.5	31.9 ± 1.0	*A*	
Dohad	*ba*	W.R.	1.493	10.041	72.6	38.3 ± 1.2	*A*	
Dhandhuka	*ba*	W.R.	1.153	13.133	76.3	64.3 ± 2.0	*A*	
Dhandhuka	*ba*	W.R.	0.513	9.733	65.4	104.2 ± 3.3	*A*	
Pavagarh	*ba*	W.R.	0.373	4.289	56.7	65.1 ± 2.1	*A*	
Saurashtra	*dy*	W.R.	1.034	8.577	51.9	47.2 ± 1.5	*A*	
Pachmari	*dy*	W.R.	0.493	5.489	67.2	62.9 ± 2.0	*A*	
		Balasubrahmanyan and Snelling [1981]						
Sagar	*ba*	W.R.	0.495	5.512	61			
				6.389	60	67.6 ± 2.0	*A*	
Sagar	*ba*	Pl	0.162	2.555	59			
				2.701	64	92.4 ± 3.0	*B*	Ar excess
Chindwara	*ba*	W.R.	0.672	8.726	80			
				8.895	71			
				9.515	80	74.9 ± 2.0	*A*	
Chindwara	*ba*	Pl	0.411	7.979	51	108.5 ± 3.0	*B*	Ar excess
Chindwara	*ba*	Py	0.136	0.604	4	25.4 ± 5.0	*C*	
Bombay	*ba*	W.R.	0.234	4.041	54			
				4.013	40			
				5.211	62	112.1 ± 3.0	*B*	Ar excess
Bombay	*ba*	Pl	0.195	3.080	22	88.8 ± 4.0	*B*	Ar excess
Annavi	*ba*	W.R.	1.89	21.350	72	63.9 ± 2.0	*A*	
Annavi	*ba*	Py	0.058	0.287	11	28.2 ± 5.0	*C*	
Saurashtra	*dy*	W.R.	0.317	4.962	51			
				4.877	40			
				5.046	42			
				4.084	46	80.8 ± 3.0	*B*	Ar excess
Saurashtra	*dy*	Pl	0.195	3.794	52	108.7 ± 4.0	*B*	Ar excess
Saurashtra	*dy*	Py	0.060	0.221	5	21.1 ± 4.0	*C*	
Saurashtra	*dy*	W.R.	0.412	4.601	37	63.6 ± 2.0	*A*	
Saurashtra	*dy*	Pl	0.717	7.472	30	59.0 ± 2.0	*A*	
Saurashtra	*dy*	Py	0.048	0.242	10	28.8 ± 5.0	*C*	
Pachmari	*dy*	W.R.	0.619	5.468	52	50.1 ± 2.0	*B*	Ar loss
Pachmari	*dy*	Pl	0.375	4.766	29	71.8 ± 4.0	*A*	
Pachmari	*dy*	Py	0.105	0.717	18	38.9 ± 2.0	*C*	
Girnar	*ns*	Hb	0.351	3.540	36	57.8 ± 2.0	*A*	
Girnar	*ad*	Hb	1.48	17.899	51	68.3 ± 2.0	*A*	

Corrected: the ages have been recalculated with new decay constants [*Steiger and Jäger*, 1977]; $\lambda_\varepsilon = 0.581 \times 10^{-10}$; $\lambda_\beta = 4.962 \times 10^{-10}$; ^{40}K/total $K = 1.167 \times 10^{-4}$. Rock: petrologic family of the studied rock; *ab*, alkaline basalt; *ad*, amphibole diorite; *an*, ankaramite; *ba*, basalt (tholeiite); *di*, diorite; *dy*, dyke; *mu*, mugearite; *ns*, nepheline syenite; *oh*, olivine nephelinite; *rh*, rhyolite; *tr*, trachyte. Mineral: mineral used for analyses; *W.R.*, whole rock; *Hb*, hornblende; *Pl*, plagioclase; *Py*, pyroxene. *K* is given in weight percent; rad^{40}Ar is given in 10^{-11} moles/g. Age: uncertainties are given at the 1σ (68%) confidence level. Quality of dating based on author information: *A*, no observed alteration (or no alteration indication given by the authors); *B*, alteration observed on bulk samples or on thin sections or deduced from chemical analyses; *C*, datings obtained on pyroxene and used as indications of Ar motion in the rocks (these ages are not used in Figure 2*b* and related discussions).

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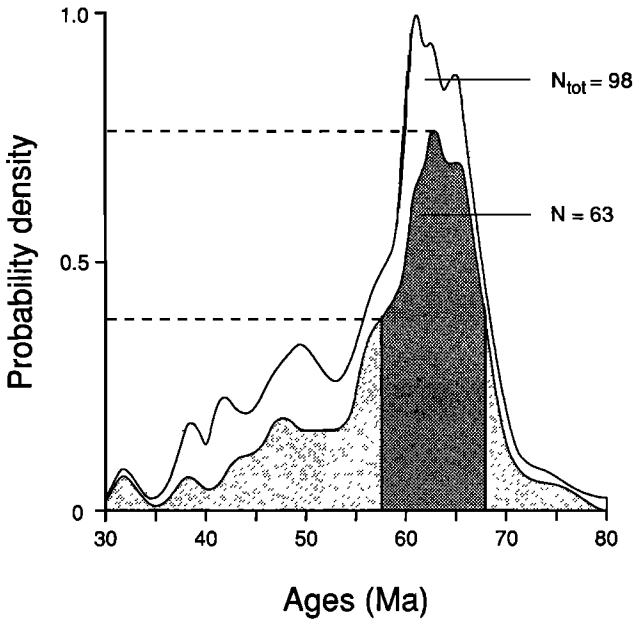


Figure 4. Summation of the three histograms of Figure 3 (all sample petrologies mixed). The light shaded area corresponds to the best data (A quality in Table 2). The darker shaded area is bounded by the ages at half amplitude of the A quality data peak and corresponds to the K-Ar age bracket retained for Deccan volcanism. N_{tot}: total number of measurements (A and B qualities); N: number of A quality measurements.

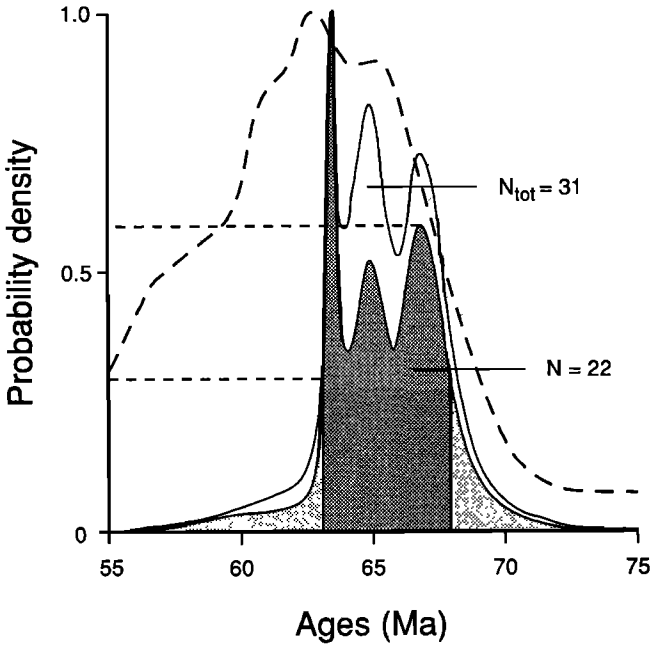


Figure 5. Same as Figure 2 with all ^{40}Ar–^{39}Ar plateau ages reviewed in this paper and listed in Table 3. The light shaded area corresponds to the best data (A quality in Table 3). The darker shaded area is bounded by the ages at half amplitude of the more significant peak of A quality data and corresponds to the ^{40}Ar–^{39}Ar age bracket retained for Deccan volcanism. The dashed line recalls the shape of the best-quality K-Ar age histogram (Figure 4). N_{tot}: total number of measurements (A and B qualities); N: number of A quality measurements.

bias of K-Ar ages resulting from argon loss. The complicated structure of this peak is likely to be an artifact without real significance. We did not find any correlation between the narrow peaks and either sampling location, or laboratory, or monitor sample. The narrow and high peak between 63 and 64 Ma is due to two apparently very precise ages at 63.6 Ma from *Courtillot et al.* [1988].

The shape of this histogram is somewhat different from those published successively by *Courtillot et al.* [1988] and *Jaeger et al.* [1989] for several reasons. First, the data set increased from one paper to the other: *Jaeger et al.* [1989] added the measurements of *Shaw et al.* [1990] and M. J. Kunk (personal communication, 1988) to those listed in the work by *Courtillot et al.* [1988]. In the present paper we have added the results of *Pande et al.* [1988]. Also, the two very precise ages which cause the younger peak were removed from the two previous papers on account of their unusual precision and the fact that they seemed to be outliers when compared to the overall shape of the diagram. Finally, the ages from *Duncan and Pyle* [1988] which were used in these diagrams were preliminary results (obtained at the preprint level). In the present paper we use of course the final published results and present the histogram of the complete data set, keeping in mind that the number of ^{40}Ar–^{39}Ar ages is still rather small (31 data altogether, with 22 of *A* quality, i.e., about the third of available K-Ar ages); as a consequence, the detailed shape of the histogram is probably not significant.

3.2. Paleontological Constraints

Several paleontological observations help in constraining the age of the Deccan traps. These observations are described by *Jaeger et al.* [1989] and are summarized in Figure 6. Remains of the ray Igdabatis were found by H. Cappetta (in the work by *Courtillot et al.* [1986*b*]) near Jabalpur in the Lameta beds. This ray is found in the Maestrichtian fauna from Niger [*Cappetta*, 1972]. Paleomagnetic and radiometric samples DK01 and DK03 come from a single flow near Dongargaon, 20 km to the south of Pisdura, which is intercalated within the "infratrappean" sediments and is consequently dated in the Maestrichtian.

Wells drilled off the western coast of India show the presence of Maestrichtian marine deposits below the Deccan traps and Paleocene deposits above them. The Narsapur 1 oil well in the Godavari delta, south of Rajahmundry, was drilled through the three flows to which the traps are reduced in this area and also through infratrap, intertrap, and posttrap sediments. *Govindan* [1981] has shown from a study of the planktonic assemblage that the infratrappean sediments belong to the *Abathomphalus mayaroensis* zone, in the uppermost Maestrichtian; the posttrappean sediments belong to the planktonic foraminifera zone P2, in the lower Paleocene, the oldest zone found anywhere above the traps. Likewise, at the opposite

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TABLE 3. Summary of ^{40}Ar–^{39}Ar Age Determinations

Locality	Rock	Mineral	Total Fusion Age, Ma	$(^{40}Ar/^{36}Ar)i$	Isochron Age, Ma	$\%^{39}Ar$	Plateau Age, Ma	Quality	
Kaneoka [1980] (*Bern 4M Muscovite*)									
Ighatpuri	*ba*	W.R.	83.9	230 ±10	91.1 ± 3.8	62	≈82	*B*	(1)
Ighatpuri	*ba*	W.R.	60.9	294 ± 5	63.7 ± 2.3	85	63.2 ± 2.3	*A*	(1)
Bombay	*on*	W.R.	88.0	290 ± 14	72.0 ± 6.9		*C*		
Bombay	*ba*	W.R.	82.2	321 ± 29	74.1 ± 3.3		*C*		
Girnar	*dy*	W.R.	64.2	310 ± 10	67.2 ± 2.4	68	68.6 ± 2.4	*A*	(1)
Mahabaleshwar	*ba*	W.R.	63.1				*C*		
Mahabaleshwar	*ba*	W.R.	64.4	304 ± 7	49.1 ± 3.0		*C*		
Venkatesan et al. [1986] (*Bern 4M Muscovite*)									
Bhor Ghat	*ba*	W.R.	67.1 ± 3.8	180 ± 15	72.5 ± 4.5	71	*C*		(1)
Bhor Ghat	*ba*	W.R.	63.9 ± 4.5	25 ± 110	67. ± 4.	87	*C*		(1)
Nakhatrana	*ab*	W.R.	54.3 ± 3.5	140 ± 45	68.2 ± 4.5	76	*C*		(1)
Duncan and Pyle [1988] (*MM-hb-1 Hornblende*)									
(Mahabaleshwar)	*ba*	W.R.	57.3 ± 1.3	305 ± 11	66.9 ± 1.0	89	68.5 ± 1.2	*A*	
(Mahabaleshwar)	*ba*	W.R.	61.2 ± 0.4	289 ± 19	66.7 ± 1.0	65	64.7 ± 1.4	*A*	
(Ambenali)	*ba*	W.R.	67.0 ± 0.6	286 ± 12	67.7 ± 3.0	94	64.4 ± 0.9	*A*	
(Ambenali)	*ba*	W.R.	68.1 ± 0.6	289 ± 15	67.7 ± 1.5	76	66.6 ± 1.0	*A*	
(Thakurdavi)	*ba*	W.R.	64.1 ± 0.7	299 ± 3	66.7 ± 1.0	67	67.8 ± 1.8	*A*	
(Neral)	*ba*	Pl	59.1 ± 0.8	295 ± 2	62.1 ± 1.2	63	60.8 ± 2.1	*A*	
(Neral)	*ba*	W.R.	68.9 ± 0.6	295 ± 2	66.9 ± 0.6	70	66.8 ± 0.7	*A*	
(Ighatpuri)	*ba*	Pl	59.7 ± 2.6	288 ± 4	66.6 ± 2.0	60	62.7 ± 2.1	*B*	
(Jarwhar)	*ba*	W.R.	67.6 ± 0.6	279 ± 20	68.5 ± 0.6	91	67.5 ± 0.6	*A*	
Courtillot et al. [1988] (*Caplongue Hornblende*)									
Dongargaon	*ba*	Pl	66.5 1.0			93	66.4 ± 1.0	*A*	
Courtillot et al. [1988] (LP6 biotite)									
Banganga	*ba*	W.R.	61.0 ± 0.2			69	63.6 ± 0.1	*A*	
Khalghat	*dy*	Pl	64.1 ± 0.3			80	63.6 ± 0.3	*A*	
Barwah	*ba*	Pl	69.8 ± 0.8			71	67.6 ± 0.9	*A*	
Barwah	*ba*	Pl	65.1 ± 0.3			98	65.1 ± 0.3	*A*	
Shaw et al. [1990] (*Unknown Monitor Sample*)									
Narmada	*ba*	W.R.					67.1 ± 1.1	*A*	(2)
Mahabaleshwar	*ba*	W.R.					67.4 ± 0.4	*A*	(2)
M. J. Kunk (personal communication 1988) (Preliminary Results) (Unknown Monitor Sample)									
Mahabaleshwar	*ba*	W.R.		271 ± 14	66.7 ± 1.3		64.4 ± 0.7	*B*	
Ighatpuri	*ba*	W.R.		264 ± 12	<66		65.4 ± 1.0	*B*	
Ighatpuri	*ba*	W.R.		276 ± 11	66.5 ± 1.2		64.5 ± 1.4	*B*	
Ighatpuri	*ba*	W.R.		218 ± 35	<67		64.6 ± 0.7	*B*	
Ighatpuri	*ba*	W.R.		345 ± 115	64.8 ± 1.9		65.3 ± 0.7	*A*	
Rajamundry	*ba*	W.R.		302 ± 5	63.7 ± 0.8		64.1 ± 0.7	*A*	
Rajpipla	*ba*	W.R.	61.4 ± 0.7				*C*		
Pande et al. [1988] (*Minnesota Hornblende*)									
Majal	*ba*	W.R.	67.0 ± 0.6	286 ± 19	67.4 ± 1.7	72	66.8 ± 0.3	*A*	
Bhuj	*ab*	W.R.	66.8 ± 1.0	299 ± 4	64.1 ± 1.0	99	65.3 ± 0.7	*A*	
Bhuj	*ab*	W.R.	63.5 ± 0.4	312 ± 6	62.1 ± 1.0	31	66.6 ± 0.9	*B*	(3)
Dharam	*ab*	W.R.	67.8 ± 1.2	330 ± 6	65.7 ± 1.4	32	67.7 ± 0.7	*B*	(3)
Vithon	*ab*	W.R.	63.7 ± 0.5	263 ± 71	65.3 ± 1.8	86	64.4 ± 0.6	*A*	
Nukhatrana	*ab*	W.R.	66.0 ± 0.5	269 ± 96	64.3 ± 1.7	62	65.1 ± 0.7	*B*	(4)

See explanations for Table 2. Monitor samples are given after the reference. Localities for *Duncan and Pyle* [1988] given between parentheses correspond to basalt formations as defined by *Beane et al.* [1986]. Quality: the alteration of *B* quality data has always been identified by an initial ratio $(^{40}Ar/^{36}Ar)i$ different from the atmospheric one. *C* quality data do not provide plateau ages. (1) Plateau and isochron ages and initial ratios $(^{40}Ar/^{36}Ar)i$ recalculated by *Baksi* [1987, 1988]; (2) no quality indication given by the authors; (3) isochron ages calculated by linear regression of 600°C–1000°C fraction, and (4) plateau age calculated for low-temperature phase only.

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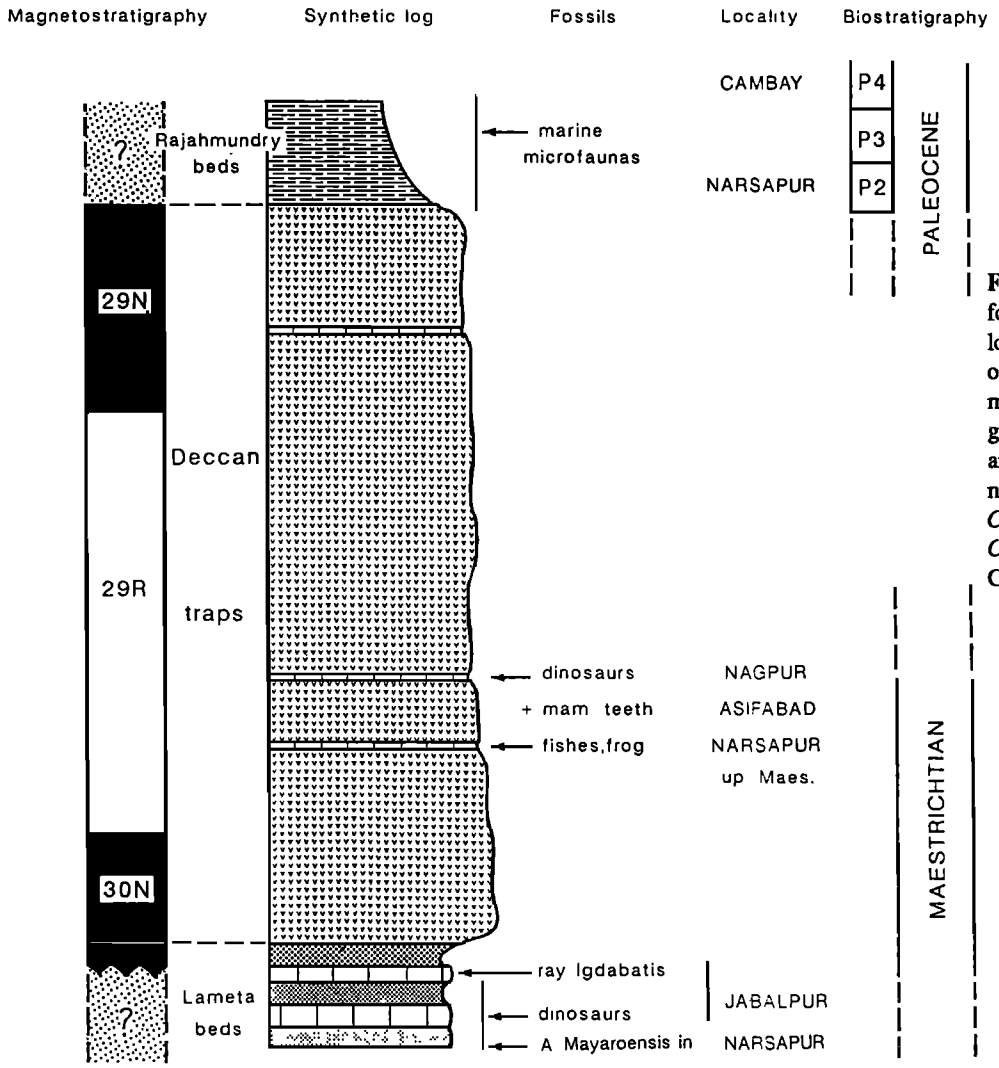


Figure 6. Schematic type section for the Deccan Traps (synthetic log), together with the underlying, overlying, and intertrappean sediments, summary of biostratigraphic data with their provenance and age determination, and magnetostratigraphic interpretation of *Courtillot et al.* [1986*b*] [after *Courtillot et al.*, 1990] (see also Courtillot [1990]).

side of the Deccan, in the Cambay basin, the marine sediments overlying volcaniclastics which are also above the traps belong to the P4 planktonic foraminifera zone [*Ramanathan*, 1981].

All available paleontological data show that volcanism started within the Maestrichtian, probably within the *A. mayaroensis* zone, and ended during the Paleocene, prior to or within the P2 zone. More observations from the other parts of the Deccan, particularly the intertrappean levels, would clearly be desirable. Unfortunately, there are few occurrences of these levels, and they are often thermally altered by the flows in such a way as to prevent fossil identification (J.-J. Jaeger, personal communication, 1987).

3.3. Deccan Age

The comparison between paleontological and radiometric data requires at least a good calibration age for the Cretaceous-Tertiary boundary (KTB) and a correct interpolation of the sea-floor magnetic signal to estimate the duration of biostratigraphic periods above and below the KTB. The estimate of the KTB age in reviews of age determinations or in magnetostratigraphic time scales, since it was first correlated with chron 29R [*Alvarez et al.*, 1977; *Lowrie and Alvarez*, 1977], varies from 63.0 to 66.7 Ma and is based on only a few studies.

The KTB age of *Van Hinte* [1976], who reviewed all preexisting numerical time scales, was at first considered as a tie point at 65 Ma in the magnetic polarity time scale of *LaBrecque et al.* [1977], then corrected to 66.7 Ma with the revised decay constants for K-Ar [*Steiger and Jäger*, 1977; *Dalrymple*, 1979] by *Ness et al.* [1980]. It should be noted that *Van Hinte* [1976] retained a KTB age at 65 Ma in his original paper, but considered that "the correct age, though, very well could be −64 rather than −65 Ma generally used" (i.e., −65.6 rather than −66.7 Ma after correction for the new constants, with an illusion of improved accuracy due to the inclusion of a figure after the decimal point). During the same period, *Armstrong* [1978] reviewed a great number of radiometric ages and estimated the KTB age at 65 Ma, after decay constant correction. *Lamphere and Jones* [1978], *Odin* [1978], and, some time later, *Curry and Odin* [1982] obtained the same estimate

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based on selected data, mainly from the American continent. *Harland et al.* [1982] revised the magnetostratigraphic time scale using the age of *Armstrong* [1978] and *Lamphere and Jones* [1978], in order to minimize seafloor spreading accelerations.

Berggren et al. [1985] used another approach to determine the KTB age. They divided the late Cretaceous and Cenozoic time scale into three segments: after Anomaly (An) 5, betweeen An 5 and An 24, and before An 24. The last period which is of interest to us is defined by interpolations between An 24 and An 34 and provides an estimate of the KTB age at 66.4 Ma. The age of An 34 (Santonian-Campanian boundary) is dated at 84 Ma according to the commonly used estimate of *Obradovitch and Coban* [1975]. On the other hand, the age of An 24 is linearly extrapolated from the second segment which is calibrated with An 5 and with three tie points (An 12, 13, and 21) on the basis of a magnetostratigraphy [*Prothero et al.*, 1982] whose interpretation is not unique and on age determinations wthout published uncertainties (for a precise analysis of this time scale, see *Lowrie* [1989]). Therefore the age estimate of An 24, and subsequently the part of the time scale prior to An 24, and thus the KTB age itself, all seem rather unreliable.

Recently, studies of bentonites from North American sections straddling the KTB with different dating methods (K-Ar, Rb-Sr, and U-Pb) have yielded new estimates of KTB age: 66 Ma for *Obradovitch et al.* [1986] and 64.4 ± 1.2 Ma for *Baadsggaard et al.* [1988]. These two studies therefore yield the lowermost (63.2 Ma) and uppermost (66 Ma) KTB age limits, if we consider the corrected age of *Van Hinte* [1976] at 66.7 Ma as possibly excessive and the interpolated age of *Berggren et al.* [1985] at 66.4 Ma as doubtful. Finally, the *Curry and Odin* [1982] estimate of 65 +1/−1.5 Ma is in agreement with most published studies. Consequently, we need a stratigraphic time scale which places the KTB at 65 Ma, to compare the paleontological, paleomagnetic, and radiometric data reviewed in this paper. We therefore choose the recent geological time scale of *Harland et al.* [1989], which retains their first age determination of the KTB at 65 Ma [*Harland et al.*, 1982], in spite of the chronogram age at 66 Ma that they obtain from Danian and Maestrichtian determinations.

Paleontological data would indicate that Deccan volcanism started in the Maestrichtian, after 74 Ma, possibly (locally) in the *A. mayaroensis* zone around 67.5 Ma, and probably finished prior to the P2 zone, i.e., before 60.5 Ma. These constraints imply that suggestions of a major eruptive phase before 75 Ma [*Alexander*, 1981; *Rao et al.*, 1985] or after 60 Ma [*Pal*, 1969; *Pal and Bhimasankaram*, 1971*c*; *Alexander*, 1981] cannot be considered as valid anymore. Furthermore, Deccan volcanism lasted only a short time as already suspected (about 5 m.y.) by *McElhinny* [1968]. More recently, we have reemphasized that the small number of reversals found in the Deccan was compatible with most of the volcanism being concentrated over a ≈1 m.y. long period straddling the KTB (anomaly 29R) (*Courtillot et al.* [1986*a*, *b*]; see also *Gallet et al.* [1989] and *Jaeger et al.*, [1989]). If this interpretation is correct, the K-Ar age histogram (Figure 4) should be interpreted in terms of a convolution between a narrow true age distribution and a "filter" characterizing alteration and argon loss. In that case the precise location of the peak and duration of the tail would not reflect volcanic activity but primarily parameters linked to the argon-loss process, petrologic lava type, climate, and weathering. This would account for the asymmetric nature of the histogram with its sharp rise, actually the most significant feature with respect to true age. This magnetostratigraphic interpretation would imply that Deccan volcanism was essentially over by 64 Ma, and in turn that all ages on the younger side of the histogram peak have been rejuvenated through some alteration process.

On the other hand, the ^{40}Ar–^{39}Ar method, which allows one to better identify and discard samples with signs of alteration, provides a symmetrical age spectrum (between 68 and 63 Ma) which is relatively accurately centered on the age range of the KTB discussed above. *Mahoney* [1988] also argues for the brevity of Deccan volcanism, on the basis of the virtual absence of intertrappean sedimentary layers in many of the thickest sections of the western traps (see also *Karmakar* [1978]) and the lack of erosional profiles developed between successive eruptions. Actually, some intertrappean layers can be observed, but they are mainly composed of ashes or lake sediments which are both characterized by of high sedimentation rate and thus correspond to short periods of time. Finally, *Jaeger et al.* [1989] point out that the overall mean of the best ^{40}Ar–^{39}Ar ages for the Deccan may yield one of the best estimates of the KTB age. With date available to them they find this age to be 65.7 ± 2.0 Ma, which we correct to 65.5 ± 2.5 Ma with our updated data set (*A* quality, Figure 5). Indeed, this value compares very favorably with the independent KTB age range at 65 + 1/−1.5 Ma [*Curry and Odin*, 1982].

4. PALEOMAGNETIC RESULTS

4.1. New Results

This section contains new and updated paleomagnetic results.

4.1.1. Sampling, measurements, demagnetization. Twenty-seven sites were sampled along the road from Dongargaon to Nagpur, then to Bombay and finally Poona (Figure 1). Preliminary results were given by *Courtillot et al.* [1986*a*, *b*] and are updated and complemented here. In each site, 4 to 10 cores were drilled and oriented with a magnetic, and as often as possible a solar compass. Seventeen sites come from single flows; three flows are sampled at more than one site (sites DK01 and 03 from a flow in Dongargaon; DK04 to 07, and DK08, 09, 11 respectively from two flows in Nagpur). All these flows

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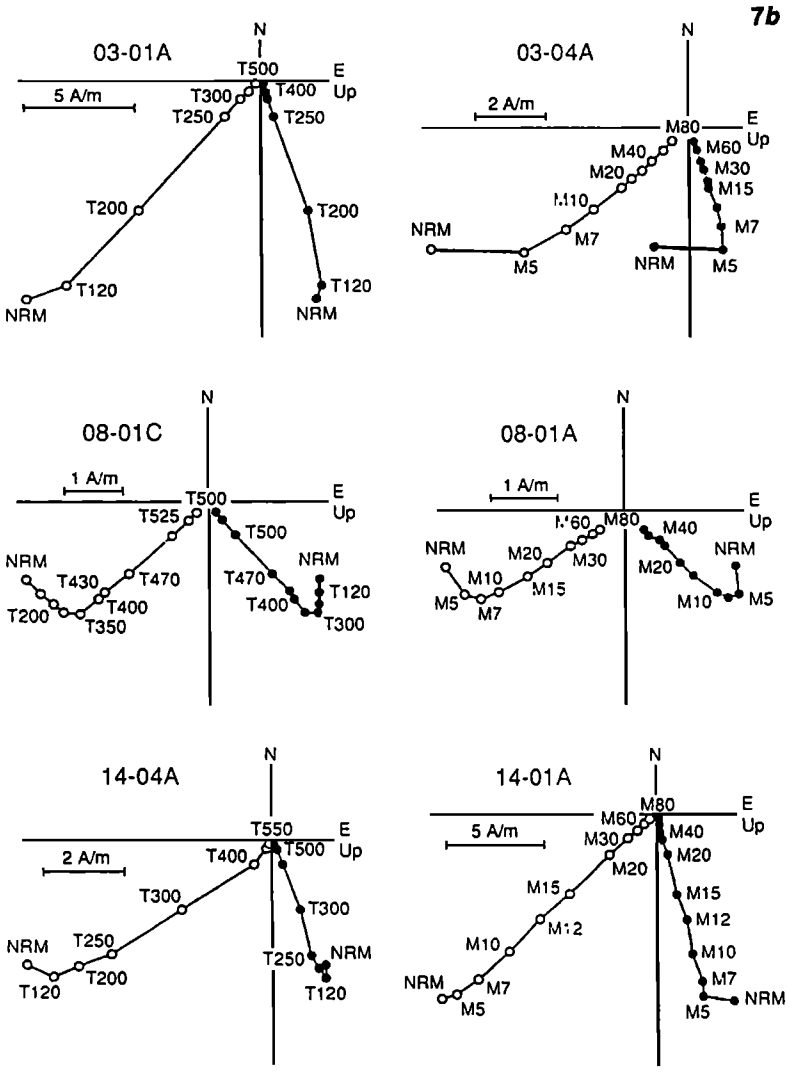


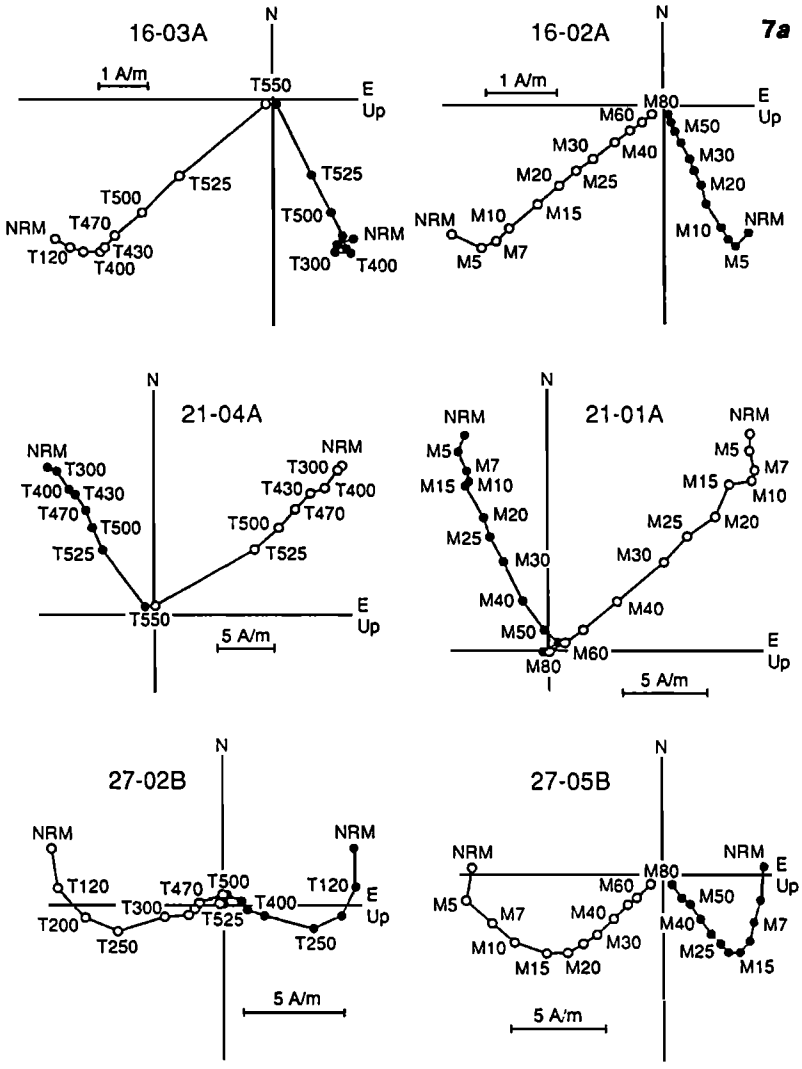
Figure 7. Typical examples of Zijderveld (orthogonal projection of the magnetization vector on the horizontal and meridian vertical planes during demagnetization) diagrams from six sites. In each site, both a thermal (left) and an alternating field (right) demagnetization are shown. Open and full symbols indicate projections on the N-S vertical plane and the horizontal plane respectively. Stratigraphic dip was negligible at these sites. (See opposing page for *7b*.)

are basaltic, except for one trachytic occurrence (DK23). One site (DK10) is actually from sediments, namely the Takli intertrap in Nagpur. The nature and distribution of sites was primarily designed to provide us with an overall view of the magnetic properties of Deccan lavas and at the same time allow a first hand critical evaluation of the published data set.

Each core was cut into at least two specimens, which were respectively demagnetized by stepwise heating in a laboratory build "zero field" furnace and by an alternating field (AF) with a Schonstedt DSM1 instrument. Magnetic measurements were performed either on a Minispin (Molspin) spinner magnetometer or a three-axis CTF cryogenic magnetometer. Inspection of natural remanent magnetization (NRM) directions shows that they are somewhat scattered, although two polarities are already apparent, with half of the NRM directions grouped in the lower southeast quadrant. The mean NRM intensity is 3.4 Am^{-1}, with the intensity histogram following a lognormal distribution. Values range from 0.1 to 10 Am^{-1}. The demagnetization data were plotted on vector (Zijderveld) diagrams, and characteristic directions were determined as line or plane data using principal component analysis [*Kirschvink*, 1980]. Flow averages were determined using *Fisher* [1953] statistics or by intersection of remagnetization circles [*Halls*, 1978].

A representative set of demagnetization diagrams is shown in Figure 7. In many cases, both the AF and the thermal technique uncovered a dominant characteristic component, with only minor secondary overprints which were generally erased before 200°C or 5 mT (Figure 7; DK03 and 14). In a number of instances the secondary component was larger, with an unblocking temperature spectrum up to 350°–400°C or a coercivity spectrum up to 15–20 mT (Figure 7; DK08, 16). In most cases the characteristic, linear component is identical with both demagnetization techniques, except in some sites such as DK27 (Figure 7) where the AF method appears to be more efficient than the thermal one in removing the secondary overprint. In some sites there is an overlap of unblocking temperature or coercivity spectra over the entire experimental range. In sites DK02, 04, 05, and 15, planar demagnetization diagrams were obtained. In all cases, fortunately, the specimens shared a common, clearly primary, reversed remanence overprinted by a more recent, roughly normal but rather scattered overprint (Figure 8).

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Following *Halls* [1978], the demagnetization trajectories for each specimen from these sites were fit by great circles. The poles to these great circles were themselves arranged along a great circle, whose pole provided the common characteristic component. An example is shown in Figure 8, with eight specimens from sites DK04 and DK05, sampled from the same lava flow. In site DK06 a large overprint was obvious on AF diagrams; the high coercivity component was not free from this overprint (specimen DK06-04A; Figure 9) and the corresponding thermal diagrams were very noisy, possibly due to some remagnetization during heating (specimen DK06-02A; Figure 9). Good results were obtained by successively applying a heating step up to 120°C, which successfully removed the large overprint, followed by AF cleaning which uncovered the primary component (specimens DK06-01B and DK06-06A; Figure 9). This composite cleaning was applied to most specimens from site DK06.

Maximum unblocking temperatures around 550°C, median AF demagnetizing fields of ≈20 mT and NRM intensities are all consistent with some form of magnetite being the essential carrier of the characteristic magnetization.

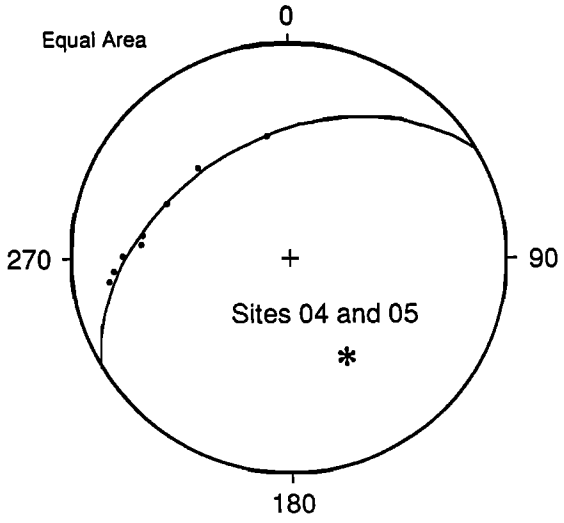


Figure 8. Example of use of the *Halls* [1978] technique of remagnetization circles (see text) for eight specimens from sites DK04 and 05, sampled in the same flow. The dots indicate poles to demagnetization planes for the eight specimens. The circle shows the trace of the least square estimate of the plane containing these poles. The star indicates the pole of this plane and consequently the best estimate of the magnetic direction common to sites DK04 and 05 and interpreted as being primary.

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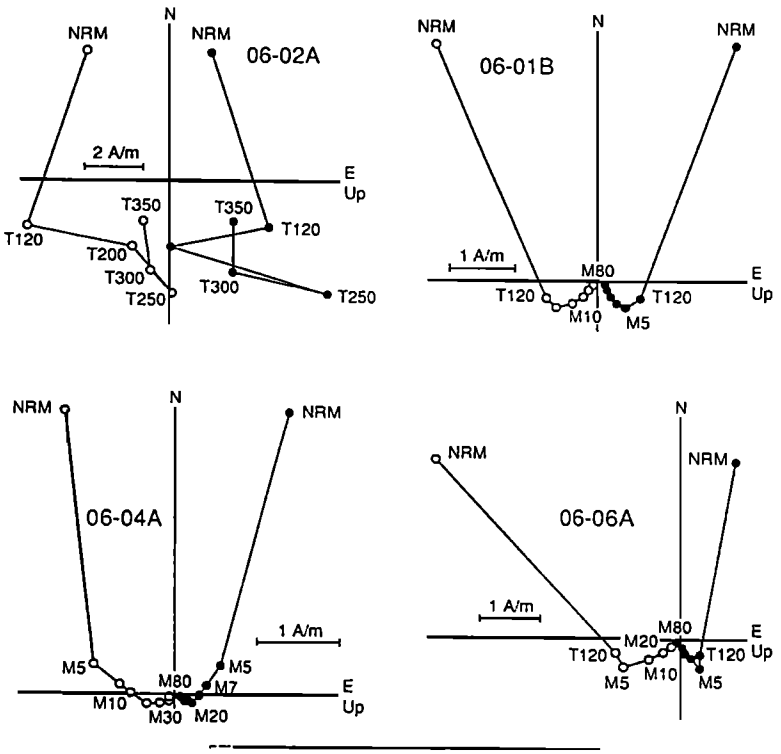


Figure 9. Typical examples of Zijderveld diagrams from site DK06. On the left-hand side, full thermal (above) and alternating field (below) demagnetizations are shown. On the right-hand side, results are shown for two specimens which have first been heated to 120°C and have then been demagnetized by alternating field. Open and full symbols indicate projections on the N-S vertical plane and the horizontal plane respectively. Stratigraphic dip was negligible at this site.

4.1.2. Statistical results. Characteristic directions for all specimens in a flow or site were averaged using Fisher statistics on directions determined by least squares (or Halls' circle technique in some cases). Information on site location and demagnetization range used in isolating the characteristic component is listed in Table 4, and site mean directions are given in Table 5. Sites coming from the same flow have been averaged, and Table 5 lists mean directions for 20 separate flows (plus the result for the Takli intertrap) and their corresponding virtual geomagnetic poles (VGPs, i.e., position of the pole of a virtual dipole that would yield the observed direction at the site). There is some amount of scatter between site mean directions within the same flow. This may be due to remaining early or late overprints, for instance due to a later reheating of the upper surfaces by succeeding flows; the number of sites per flow (two or three in three cases) is too small to allow a more elaborate quantitative analysis.

Confidence intervals (at the 95% probability level) for site mean directions range from 2° to 17°. Only three flows have an α_{95} exceeding 10°, all from the western part of our traverse (DK17, 23, 27). One of the sites studied by the Halls technique (see DK15 in Table 5) gives a large (greater than 10^{-5}) Q parameter [*Halls*, 1978], indicative of another unreliable direction. Because of the wide geographical coverage of the sites (over almost 1000 km), individual site mean directions cannot be compared directly and individual-flow VGPs must be used in the statistical analysis. The Fisherian mean pole for the 20 flows is at 37.7°N, 283.3°E with an A_{95} = 7.0° and an angular standard deviation of 17.0° (Table 6). If we choose to exclude the four flows for which $\alpha_{95} > 10°$ or $Q > 10^{-5}$, the Fisherian mean pole changes insignificantly (38.4°N,

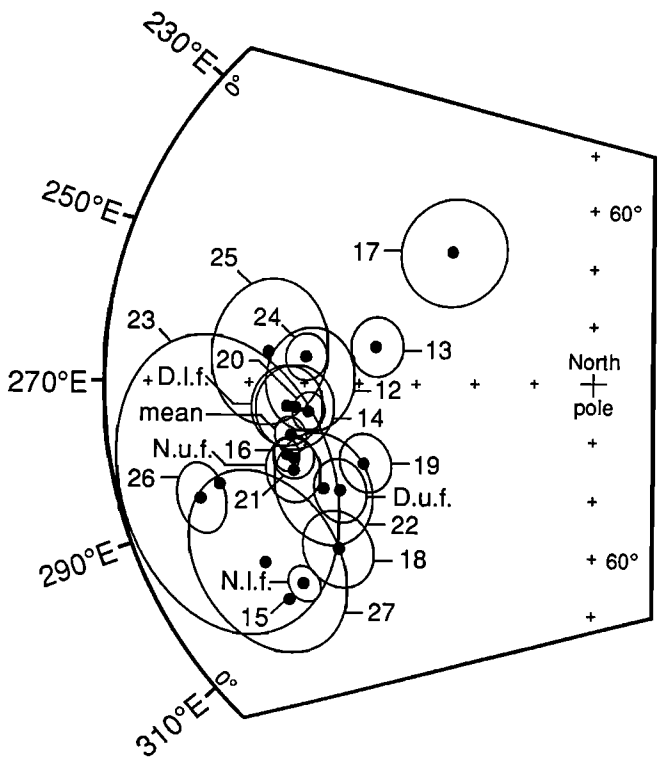


Figure 10. Virtual geomagnetic poles with 95% confidence circles for the 20 flows from the Dongargaon-Bombay traverse reported in this paper (except for the pole of site DK15 which is determined by the Halls Technique only). The shaded area indicates the 95% confidence circle of the mean poles of the 16 selected flows discussed in the text. Several sites coming from a single flow have been averaged in Dongargaon (D) and Nagpur (N) (u.f. and l.f. stand for upper and lower flow, respectively); all the other site numbers are indicated. Data are in Table 5.

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282.4°E), A_{95} is slightly smaller at 6.1° and the angular standard deviation drops to 13.2° (Table 6). The distribution of flow VGPs (Figure 10) shows some elongation in longitude. This is confirmed by a moment of inertia analysis [*Bingham*, 1974; *Mardia*, 1975] (using the algorithm of *Gillett* [1986]). The data distribution is found not to possess circular symmetry about the mean because of a too large ratio of the Bingham precision parameters along the two main eigenvectors ($K1/K2$ = 4.0); the maximum eigenvector (for N = 20) is at 37.4°N, 283.6°E (virtually identical to the Fisherian mean) with confidence intervals of 4.0° and 8.2°, the long axis trending toward 128.7°. With the reduced (N = 16) data set the pole moves to 38.5°N, 282.3°E, and the confidence intervals become 4.3° and 6.6°, with a 113.2° azimuth (Table 6). The confidence ellipse is slightly less elongated: the ratio of major to minor axis falls from 2.07 to 1.54. We note that this elongation value and its azimuth are comparable to what we might have expected if the population of remanent directions (and not poles) were itself Fisherian. Indeed, at

TABLE 4. Paleomagnetic Sites Reported in this Paper and Some Characteristics of Magnetic Measurements

Site	*Location*	*N*	*Sample Treatment*	*Data Analysis*	*Comments*
			Dongargaon, 20°31′N, 79°17′E		
01	1	9	Th(5) AF(4)	Kir. 10 mT/120°C	
02	2	9	Th(6) AF(3)	Halls	(1)
03	1	6	Th(3) AF(3)	Kir. 10 mT/120°C	
			Nagpur, 21°10′N, 79°12′E		
04–05	2	8	Th(6) AF(2)	Halls	(2)
06	2	4	conj.	Kir. 120°C + 10 mT	(2)
07	2	3	Th(2) AF(1)	Kir. 10 mT/350°C	
08	1	5	Th(4) AF(1)	Kir. 10 mT/350°C	
09	1	3	Th(2) AF(1)	Kir. 7.5 mT/300°C	
10	(*I*)	7	Th(6) AF(1)	Kir. 15 mT/120°C	
11	1	4	Th(3) AF(1)	Kir. 15 mT/350°C	(2)
			Kondhalhi, 21°09′N, 78°59′E		
12	...	5	Th(2) AF(3)	Kir. 5 mT/120°C	
			"*Amravati*," 20°58′N, 77°50′E		
13	...	6	Th(4) AF(2)	Kir. 10 mT/200°C	
			Akola, 20°40′N, 77°05′E		
14	...	4	Th(2) AF(2)	Kir. 5 mT/250°C	
			Buldana, 20°40′N, 76°18′E		
15	4	4	Th(3) AF(1)	Halls	(2)
16	7	6	Th(4) AF(2)	Kir. 5 MT/430°C	
			Ighatpury, 19°41′N, 73°38′E		
17	1	7	Th(5) AF(2)	Kir. 15 mT/350°C	
18	2	5	Th(4) AF(1)	Kir. 7.5 mT/350°C	
19	4	5	AF(5)	Kir. 15 mT	
20	7	6	Th(4) AF(2)	Kir. 10 mT/200°C	
			Bombay, 18°56′N, 72°51′E		
21	...	7	Th(4) AF(3)	Kir. 10 mT/300°C	
22	...	3	Th(2) AF(1)	Kir. 20 mT/350°C	
23	(*T*)	5	Th(4) AF(1)	Kir. 40 mT/430°C	(2)
24	...	7	Th(4) AF(3)	Kir. 20 mT/400°C	
			Karjat, 18°53′N, 73°20′E		
25	...	5	Th(3) AF(2)	Kir. 7.5 mT/250°C	
			Lonawala, 18°46′N, 73°44′E		
26	...	9	Th(5) AF(4)	Kir. 10 mT/300°C	
			Khandala, 18°44′N, 73°29′E		
27	...	5	Th(3) AF(2)	Kir. 10 mT/350°C	(2)

Site: number of the paleomagnetic site (see location on Figure 1). Location: local stratigraphic position of the sampled flow (numbers increasing upward); ..., no stratigraphic correlation; *I*, intertrappean sediments; *T*, trachyte. *N*: number of samples used in statistics. Sample treatment: number of samples demagnetized by thermal (Th) and alternating field (AF) methods; conj., heating at 120°C followed by AF demagnetization. Data analysis: statistics used to calculate mean direction. Halls, demagnetization circle method of *Halls* [1978]; Kir., least squares method of *Kirschvink* [1980] with indication of the lower bound of demagnetization. Comments are (1) two primary components defined by the method of *Halls* [1978] because of the presence of a secondary component and (2) presence of a strong secondary component.

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TABLE 5. Paleomagnetic Results

	NRM Directions					*Characteristic Directions*					*VGPs*			
S/F	*N*	*Dec.*	*Inc.*	*K*	α95	*N*	*Dec.*	*Inc.*	*K/Q*	α95	*Long.*	*Lat.*	*dp*	*dm*
01	9	170.5	53.4	72.9	5.5	9	172.0	47.4	107.7	5.0	268.5	40.4	4.2	6.5
03	7	151.6	79.3	11.9	14.3	6	155.2	53.9	34.7	11.5	282.9	30.2	11.3	16.2
D.l.f.	16	163.8	67.8	11.2	10.8	15	165.9	50.2	46.3	5.7	274.4	36.8	5.1	7.6
02A	4	66.9	29.0	7.2	26.1	4	158.1	43.6	*2.2	...	285.1	39.4	...	...
02B	5	103.0	40.9	48.3	9.0	5	151.1	28.8	*8.1	...	299.9	44.3	...	...
D.u.f.	10	88.7	38.4	10.5	13.0	9	153.9	35.4	94.5	5.3	293.5	42.5	3.5	6.1
04-05	10	36.1	52.9	7.8	13.4	8	150.6	45.7	*6.8	...	290.8	34.1	...	...
06	10	23.1	40.1	29.1	8.2	4	155.7	43.8	390.5	4.3	287.1	37.7	3.3	5.3
07	5	205.3	50.6	3.9	31.7	3	149.1	54.2	105.5	12.1	287.5	26.8	11.9	17.0
N.u.f.	25	28.5	56.3	7.0	8.3	15	154.9	46.0	60.1	4.8	286.8	35.7	3.9	6.1
08	5	122.2	50.1	42.4	9.6	5	138.8	40.5	203.2	5.4	303.6	30.3	3.9	6.5
09	3	130.9	41.4	652.7	3.2	3	136.8	37.1	2458.4	2.5	307.1	30.6	1.7	2.9
11	4	95.9	56.7	44.0	10.5	4	131.2	44.0	310.3	5.2	306.7	23.4	4.1	6.5
N.l.f.	12	124.0	52.4	12.7	12.1	12	135.8	40.9	205.3	3.0	305.6	28.0	2.2	3.6
10(*I*)	7	152.8	39.2	88.6	6.5	7	152.4	39.6	148.6	5.0	292.4	38.7	3.6	6.0
12	5	172.4	45.5	57.6	10.2	5	171.1	45.6	97.1	7.8	269.6	41.1	6.3	10.0
13	6	171.5	27.9	44.1	9.4	6	178.5	30.9	146.5	5.6	260.2	52.3	3.5	6.2
14	4	160.8	52.1	99.8	7.0	4	164.0	48.0	209.1	6.4	274.8	38.1	5.4	8.3
15	4	65.3	49.5	12.4	19.9	4	129.9	42.0	*84.6	...	306.2	23.8	...	...
16	6	144.3	52.8	337.1	3.1	6	154.9	47.2	788.7	2.4	283.6	35.4	2.0	3.1
17	12	185.4	35.4	4.4	32.6	7	194.6	19.5	29.7	11.3	226.5	57.0	6.2	11.8
18	5	122.4	43.4	34.5	10.7	5	140.2	30.4	119.3	7.0	303.7	37.0	4.4	7.8
19	13	155.1	27.4	129.6	6.8	5	156.0	27.4	189.5	5.6	290.0	48.4	3.3	6.1
20	6	161.2	47.4	382.1	2.9	6	161.4	44.7	562.2	2.8	275.8	40.6	2.3	3.6
21	7	337.3	−38.8	79.6	7.8	7	331.8	−45.4	337.9	3.3	284.7	36.6	2.7	4.2
22	3	341.9	−35.5	232.3	5.3	3	328.6	−37.1	174.8	9.4	292.3	39.8	6.4	11.0
23	5	356.3	38.8	3.6	32.9	5	319.8	−57.6	21.1	17.0	286.0	21.5	18.3	25.0
24	8	348.9	−51.4	112.7	5.7	7	349.6	−49.0	333.2	3.3	264.7	40.1	2.9	4.4
25	5	161.4	74.6	39.5	10.0	5	168.4	56.3	78.2	8.7	264.4	33.2	9.1	12.6
26	9	130.1	60.8	87.5	6.7	9	135.6	60.4	188.0	3.8	287.1	17.2	4.4	5.7
27	5	54.3	36.0	3.9	35.3	5	131.8	47.2	35.9	12.9	299.5	23.8	10.8	16.8

Data from this study, updating *Courtillot et al.* [1986*b*]. *S/F*: Site and/or flow numbers (see Table 4): D.l.f., lower flow at Dongargaon including sites 01 and 03; D.u.f., upper flow at Dongargaon (site 02); N.l.f., lower flow at Nagpur including sites 08, 09, and 11; N.u.f., upper flow at Nagpur including sites 04, 05, 06, and 07; and (*I*), intertrappean sediments. *N*: number of samples used in statistics; Dec., Inc., declination and inclination; *K*: Fisherian precision parameter; *Q*: Hall's precision parameter ($\times 10^{-6}$); the *Q* values are marked with an asterisk; α_{95}: semiangle of the cone of 95% confidence; VGPs: virtual geomagnetic poles; Lat., Long., latitude (°N) and longitude (°E) of the VGPs; and *dp*, *dm*: semiaxes of the oval of 95% confidence for the VGPs.

the paleocolatitude of the Deccan ($114° < \theta < 124°$) the elongation of the uncertainty ellipse of the mean VGP [$dm/dp = (1 + 4 \cot g^2 \theta)^{1/2}$] is around 1.5, with an azimuth between 110° and 120°). Finally, we note that all remanent directions but four, i.e., 80% of the flows which we sample across our traverse, are reversed, as was noted by *Courtillot et al.* [1986*a*, *b*].

We note a significant difference between the results presented above and the preliminary ones published in the work by *Courtillot et al.* [1986*b*]. It seems that the number of flows in the earlier study (15) was not sufficient to have averaged out secular variation. Adding or removing only a few flows alters the mean pole position by several degrees. Indeed, the angular standard deviation of VGPs obtained by *Courtillot et al.* [1986*b*] was about 12.5° and was therefore much smaller than the value of 17° that we deduce from the present compilation as resulting from paleosecular variation (see below).

4.2. Compilation of Deccan Results

The section presents a compilation of Deccan data and allows one to perform a number of statistical tests.

4.2.1. Presentation of the data. Bearing in mind the observations reported above, we have undertaken as extensive as possible a compilation of earlier paleomagnetic results from the Deccan traps. We have been able to identify 35 studies which span over 30 years (appendix); the corresponding data were obtained with rather different laboratory equipment, and different demagnetization and analysis techniques. Early studies did not always include demagnetization, or did not list sufficient information to allow one to assess the reliability of the data. Previous reviews were written by *Pal and Bhimasankaram* [1971*c*] and *Klootwijk* [1979]. Klootwijk's very thorough review does not list all available flow results but includes a number of site means (means of several flows) as calculated by original authors. In contrast, our compilation uses

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the flow as a unit, and published averages of several flows are not used.

This compilation is used in conjunction with our own data (*Courtillot et al.* [1986*b*] updated in the present paper and D. Vandamme et al., manuscript in preparation, 1991) to perform a number of statistical tests on groups of data sorted as a function of quality, location, or magnetic polarity. The scatter of the 563 mean-flow VGPs listed in our compilation is displayed in Figure 11*a*. There is of course some amount of uncertainty on the polarity of poles which are remote from either the mean pole or its antipodal direction. Good grouping around the mean is observed, as is quantitatively shown by the isodensity diagram of Figure 11*b*. This diagram shows that the data depart from circular symmetry. An apparently anomalous set of data cause the low-density contours to extend westward in pear shape to longitudes in the 220°–250° range, whereas the high-density contours are somewhat elongated, again with a pear shape but toward the northeast (NE). Only 12 poles are listed by original authors as being associated with a transitional direction. Such transitional directions can either be truly related to excursions or reversals or they may indicate overprints which have not been totally removed. Actually, we see in Figure 11*a* that many VGPs are far from the overall mean, contributing greatly to the angular standard deviation. In order to define the limits beyond which the VGPs are considered as outliers of the main distribution, we use the graphical method of *Fisher et al.* [1987]. The distribution of points as a function of their angular distance to the mean and their azimuthal distribution around the mean are plotted separately (Figure 11*c* and *d*). A Fisherian distribution (i.e., a distribution of unit vectors on a sphere, analogous to a normal distribution on a plane) [*Fisher*, 1953] would be represented by a straight line going through the origin with a slope value $1/k$ (k: Fisherian precision parameter) on the colatitude plot of Figure 11*c*.

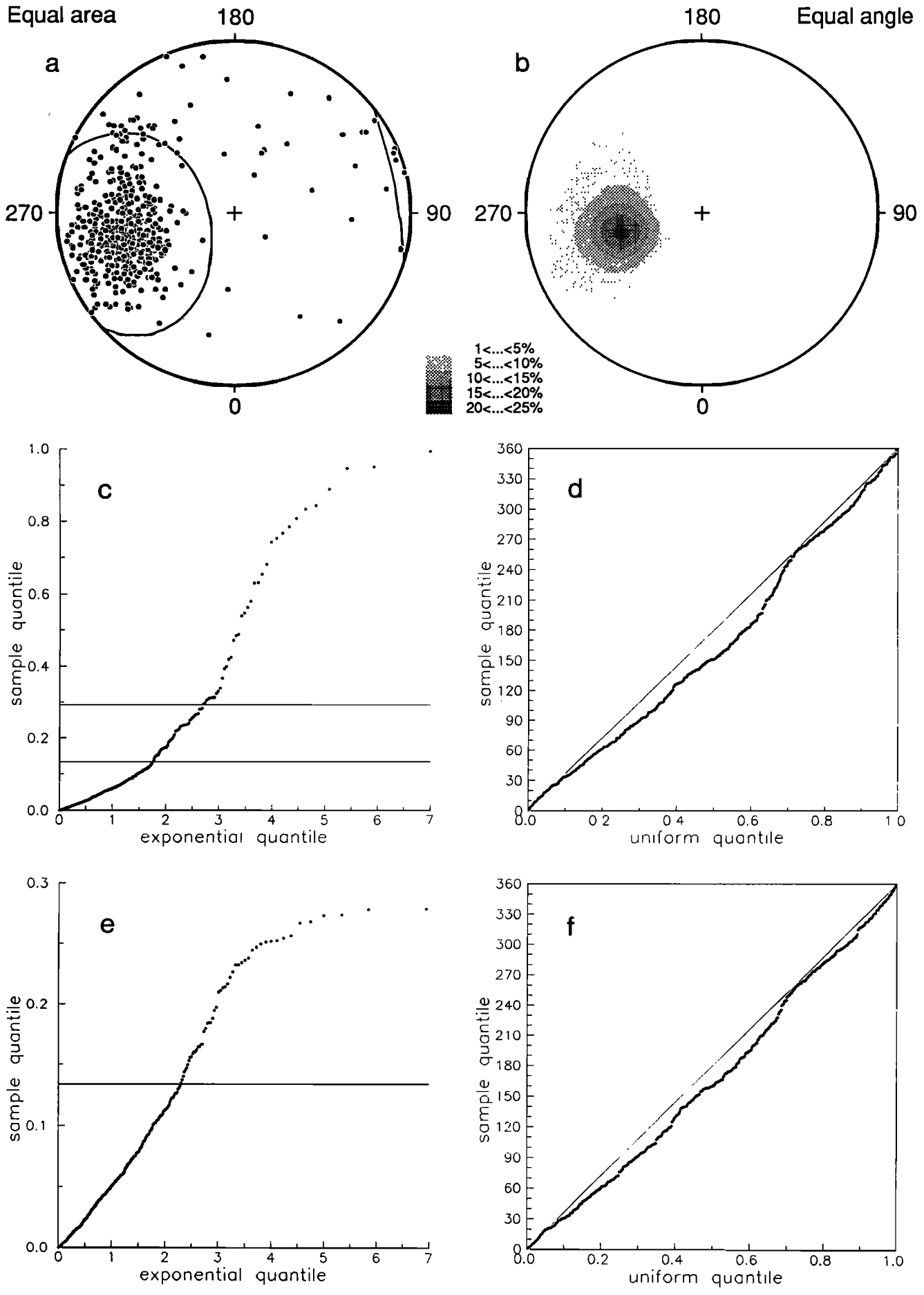
It is clear from that figure that a Fisherian distribution applies to data no further than 30° away from the mean. Another distribution with smaller k value is suggested by this diagram, extending to about 45°. Beyond this the few data points are clearly outliers; the slope of the diagram steepens, suggestive of a uniform distribution and influenced by the artificial cutoff at 90° away from the mean. The longitude plot (Figure 11*d*) shows a weak but noteworthy bow shape under the diagonal line which would characterize the expected uniform azimuthal distribution. This shape integrates the varying pear-shaped contours of the isodensity curves of Figure 11*b*; indeed, it is easy to understand the difficulties in interpreting this diagram by looking at the complex distribution highlighted by the isodensity diagram (Figure 11*b*). As a result, and for lack of a better criterion we exclude from our subsequent discussion those data points that are located more than 45° away from the mean. This relatively arbitrary selection is the same made by most authors who study paleosecular variation [e.g., *McFadden et al.*, 1988; *Merrill and McElhinny*, 1983, p. 163]; it corresponds to the decrease of magnetic intensity observed in Iceland by *Wilson et al.* [1972] and associated with the reversal process. A total of

TABLE 6. Fisher and Bingham Statistics for Various Subsets of Deccan Paleomagnetic Poles

		Fisher					*Bingham*					
Subset	*N*	*Lat.*	*Long.*	*K*	A_{95}	*s.d.*	*Lat.*	*Long.*	*A13*	*A12*	*Ratio*	*Azim.*
Present study	20	37.7	283.3	22.8	7.0	17.0	37.4	283.6	4.0	8.2	2.07	128.7
(1)	16	38.4	282.4	37.9	6.1	13.2	38.5	282.3	4.3	6.6	1.54	113.2
All poles	563	35.3	278.5	11.0	1.9	24.5	35.3	279.5	1.6	1.9	1.21	114.1
All poles <45°	519	35.5	280.3	18.3	1.5	19.0	35.6	280.6	1.3	1.6	1.18	107.2
$x_1 = 2, 3$ or 9	460	35.9	280.5	18.6	1.6	18.8	36.0	280.8	1.5	1.7	1.18	103.4
$x_1 = 2$ or 3	384	36.3	280.9	17.7	1.8	19.3	36.5	281.3	1.6	1.9	1.22	103.6
$x_1 = 3$	180	36.7	278.7	17.4	2.6	19.5	36.9	279.2	2.2	2.9	1.34	89.4
North	60	37.1	283.7	17.7	4.5	19.3	37.2	284.2	4.0	4.6	1.14	125.7
Mandla	46	36.4	287.4	31.9	3.8	14.3	36.6	287.6	2.6	4.5	1.72	20.6
Center	68	36.6	272.1	13.1	4.9	22.5	36.5	273.0	3.2	6.0	1.86	77.5
(2)	44	37.8	281.0	23.1	4.6	16.9	37.7	281.6	4.0	4.7	1.18	87.1
East	56	30.5	287.5	20.7	4.3	17.8	30.6	287.5	3.3	4.9	1.48	141.3
(3)	52	32.0	286.1	22.7	4.2	17.0	32.1	286.0	3.5	4.7	1.32	141.3
N.W. Ghats	48	38.4	281.6	22.2	4.5	17.2	38.6	281.4	3.7	4.8	1.30	85.1
Mahabaleshwar	46	39.2	277.9	19.6	4.9	18.4	39.6	277.6	4.1	5.3	1.30	162.7
S.W. Ghats	60	35.9	277.9	17.6	4.5	19.4	36.3	277.9	4.0	4.7	1.18	117.6

(1) Present study excluding the poles which correspond to flow mean directions with low precision ($\alpha_{95} > 10°$ or $Q > 10^{-5}$; see text). (2) Poles from the center region excluding the sites at Ellora and Harsul. (3) Poles from the east region excluding the site at Pohor. All poles <45°: all poles excluding those which are more than 45° away from the overall mean; x_1: quality parameter based on knowledge and value of confidence interval for the flow mean directions (see Appendix). N.W. Ghats and S. W. Ghats; respectively the northern and southern parts of the western Ghats; the boundary is at the latitutde of Mahabaleshwar; N: number of virtual geomagnetic poles (VGPs); Lat, Long.: latitude and longitude of mean VGPs; K and A_{95}: Fisherian precision parameter and semiangle of the cone of 95% confidence; s.d.: angular standard deviation; A13, A12; semiaxes of the Bingham ellipse of 95% confidence; Ratio: elongation of the Bingham ellipse of 95% confidence (A12/A13); and Azim.: azimuth of the great axis of the Bingham ellipse of 95% confidence.

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Figure 11. (opposite) Distribution of virtual geomagnetic poles (VGP) of 563 flows reviewed in this paper and mentioned in the Appendix 1. (*a*) Equal area projection of the virtual geomagnetic poles (VGP) (all plotted in the northern hemisphere for clarity) and of the circle with 45° radius centered on the overall mean and used as a selection limit (see text). (*b*) Isodensity contours of VGPs shown in Figure 11*a*. Graphs (*c*)–(*f*) Tests of the distribution using the method of *Fisher et al.* [1987]. For the colatitude plots (Figure 11*c* and 11*e*) the data are sorted as a function of their angular distance from the mean θ. The sample quantile is (1 − cos θ). The exponential quantile is the function $\log(i/N - 1)$, which results in a straight line through the origin if the distribution is Fisherian. For the longitude plots (Figures 11*d* and 11*f*) the data are sorted as a function of their azimuth with respect to the mean. The linear quantile also produces a straight line through to origin if the distribution presents circular symmetry, as in a Fisherian one. (*c*) Colatitude plot [*Fisher et al.*, 1987] of VGPs with reference lines at 30° and 45° away from the mean. (*d*) Longitude plot [*Fisher et al.*, 1987] of VGPs with reference diagonal line (uniform azimuthal distribution). Figures 11*e* and 11*f* same as Figures 11*c* and 11*d*, excluding VGPs considered as transitional by original authors and those more than 45° away from the mean.

44 points (out of 563) are thus excluded from further statistical analysis. After this winnowing of the data set, the elongation in longitude remains noticeable although the Bingham precision parameters do not permit to strictly reject circular symmetry (Table 6, result with 519 poles). The new colatitude plot (Figure 11*e*) confirms the angular point around 30°. Likewise, the longitude plot (Figure 11*f*) slightly better approaches the diagonal line, although it still bows under it. These two plots show that the data set still does not strictly conform with a Fisherian distribution.

4.2.2. Influence of data "quality." The result quality has been defined, on the basis of the α_{95} value. One hundred and eighty flow data have $\alpha_{95} < 5°$; there are 384 with $\alpha_{95} < 10°$, 460 flow data when results without published α_{95} are also accepted, and 519 when all flow data with VGPs less than 45° away from the mean are considered. We have analyzed these various data sets with both Fisherian and moment of inertia statistics [*Onstott*, 1980; *Gillett*, 1986]. The statistical results of these various tests are summarized in Table 6. The various selections (all flow data, data with α_{95} unpublished and less than 10°, data with α_{95} less than 10°, data with α_{95} less than 5° only) provide virtually identical means, with a maximum angular difference of 1.5°, a K value around 18 and an angular standard deviation around 19°. The confidence intervals vary between 1.5° and 2.6°.

We could have expected that the fast latitudinal drift of the Indian subcontinent during the Deccan eruption would show in the paleomagnetic results. The VGP cloud would then have been stretched along a roughly N-S direction, actually with an azimuth of about 340° considering the 20° counterclockwise posttrap rotation of India. This has indeed been invoked by a number of authors. Actually, the Bingham statistics yield 95% confidence ellipses of mean VGPs with roughly stable great axis azimuth around 100°. Furthermore, the elongation of this ellipse (indicated by the ratio of the major to minor axis) is about 1.2 regardless of quality selections. Consequently, it would seem reasonable to consider that we only observe here the VGP distribution expected from a Fisherian population of magnetic remanent directions, as was seen to be the case of our own data (see section 4.1.2), but the elongation of the confidence ellipse seems small. We have recalculated the magnetic directions at Bombay from these selected VGPs to check this hypothesis (Figure 12*a*). The colatitude distribution is close to an exponential one (Figure 12*c*), but Figure 12*d* shows a clear anisotropy of the direction distribution along an azimuth around 150° which is also sensitive in the isodensity diagram (Figure 12*b*). The direction of the maximum eigenvector of VGPs is extremely stable regardless of quality selection. It is always close to the Fisherian mean (no farther apart than 1°) because of the slightness of the elongation of the VGP distribution.

4.2.3. Influence of geographical location. The data have been divided into six regional groups (Figure 13*a*), with a separate analysis for the very extensively sampled site of Mahabaleshwar. This subdivision is not equal area because of the great spatial heterogeneity of paleomagnetic sites but is roughly equal sampling; it is largely ad hoc and has no geological basis. This regional study should allow one to detect possible outliers which can influence the overall statistics and to test for a possible geographic evolution of the VGPs. For this study we use only paleomagnetic data with $\alpha_{95} < 10°$, which leads to about 50 VGPs per region.

The regional statistical means are relatively well grouped (Figure 13*b*) and their 95% confidence circles, between 4° and 5° radius, are not small enough compared to VGP scatter to allow one to detect a regular motion of VGPs along either N-S or E-W directions. Yet, two poles lie significantly away from the overall mean, without intersecting its confidence circle (Figure 13*b*): they correspond to the "center" and "east" regions. We have therefore analyzed the data from these two regions in more detail to try to determine any possible bias that could result from local data. Indeed, the two regional VGP clouds are clearly elongated due to individual site results at Pohor in the "east" and Ellora and Harsul in the "center" (Figure 13*a*). The Ellora site has been studied in three papers [*Bhimasankaram and Pal*, 1968; *Pal*, 1975; *Wensink*, 1977] which yielded similar results, having apparently either recorded an excursion of the magnetic field or undergone a local remagnetization (C. T. Klootwijk, personal communication, 1990, recalls the vesicularity of the Ellora flows). The same is also observed in Harsul, as well as in Chandor (another closeby site which is not considered here because of the lack of published confi-

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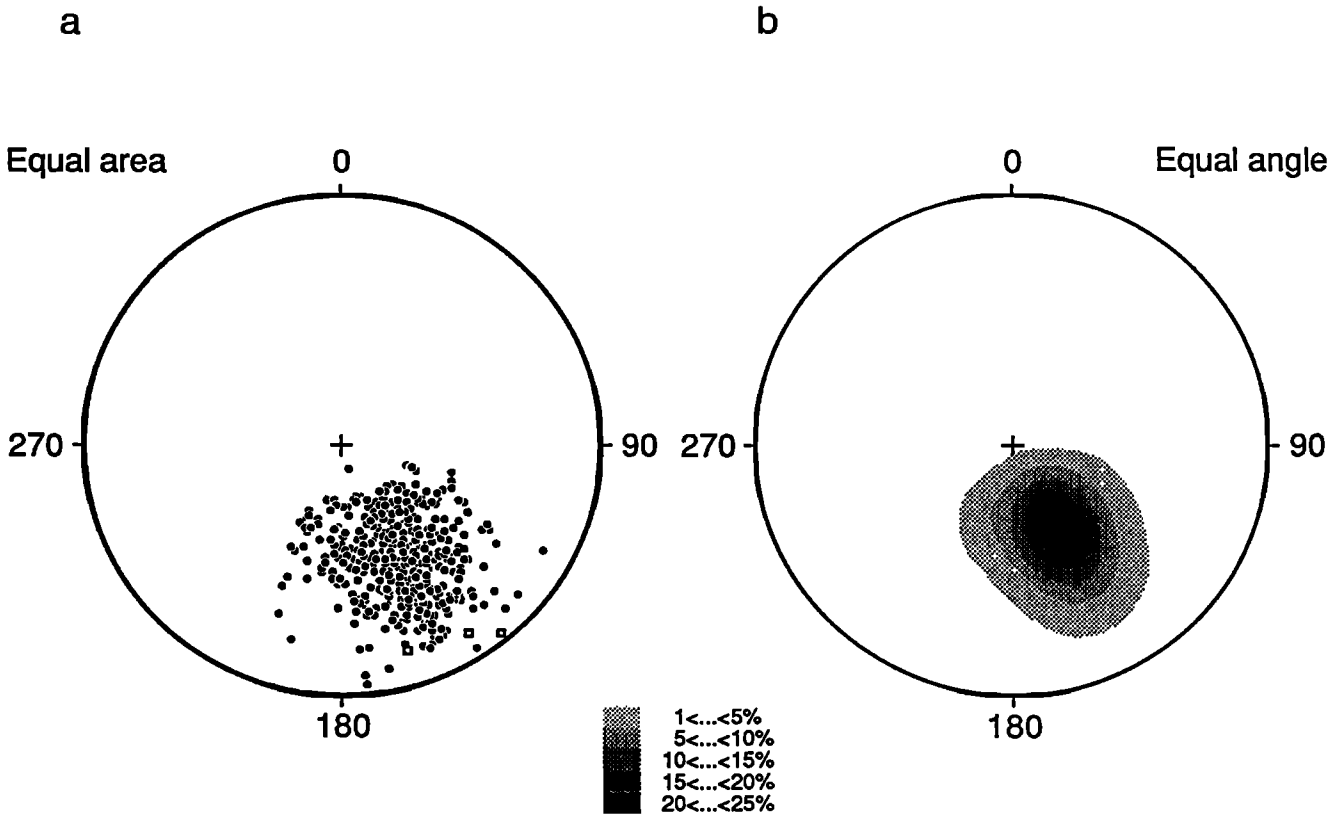
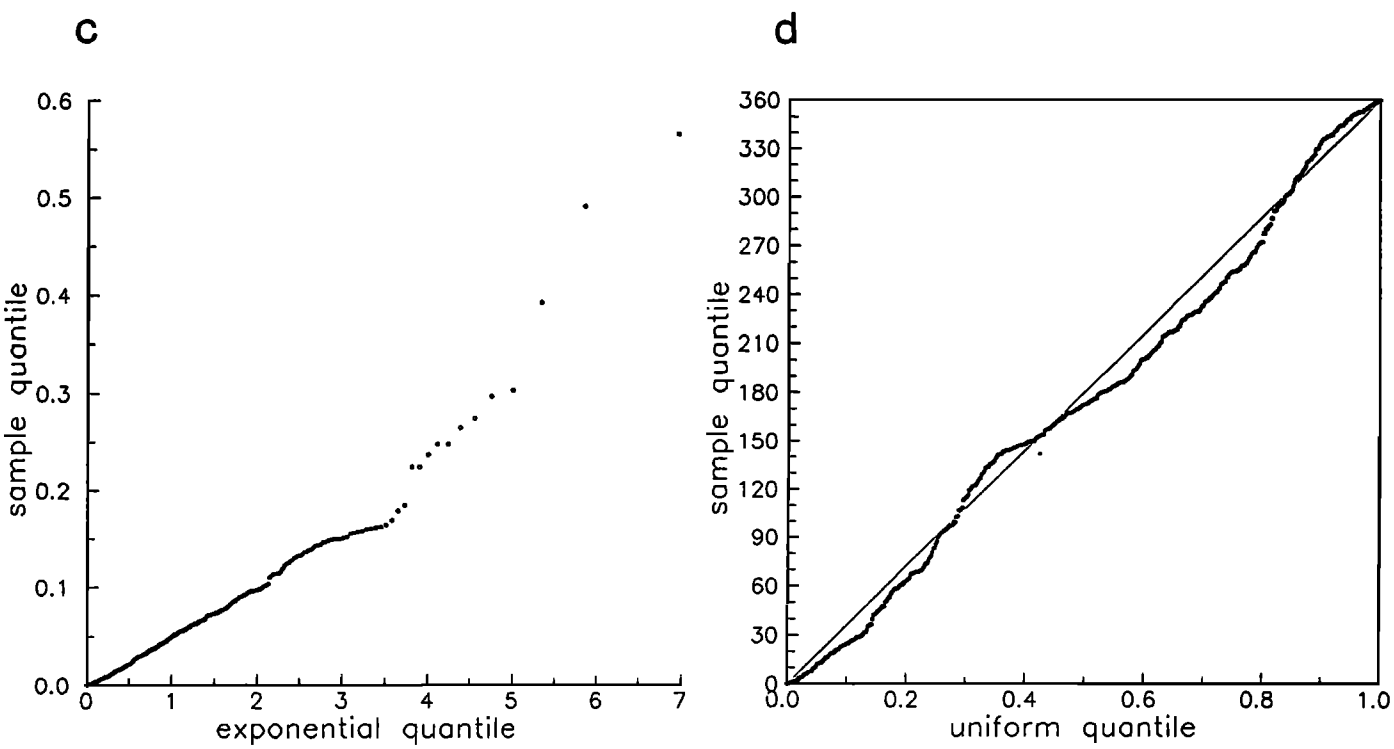


Figure 12. Same as Figure 11 but for the remanent directions, recalculated for Bombay from the virtual geomagnetic poles considered not to be transitional (see text and Figures 11*e* and 11*f*). Normal directions have been flipped to their reversed equivalent. (*a*) Equal area projection. (*b*) Isodensity diagram. (*c*) and (*d*) "Colatitude" and "longitude" plots, respectively [*Fisher et al.*, 1987]).



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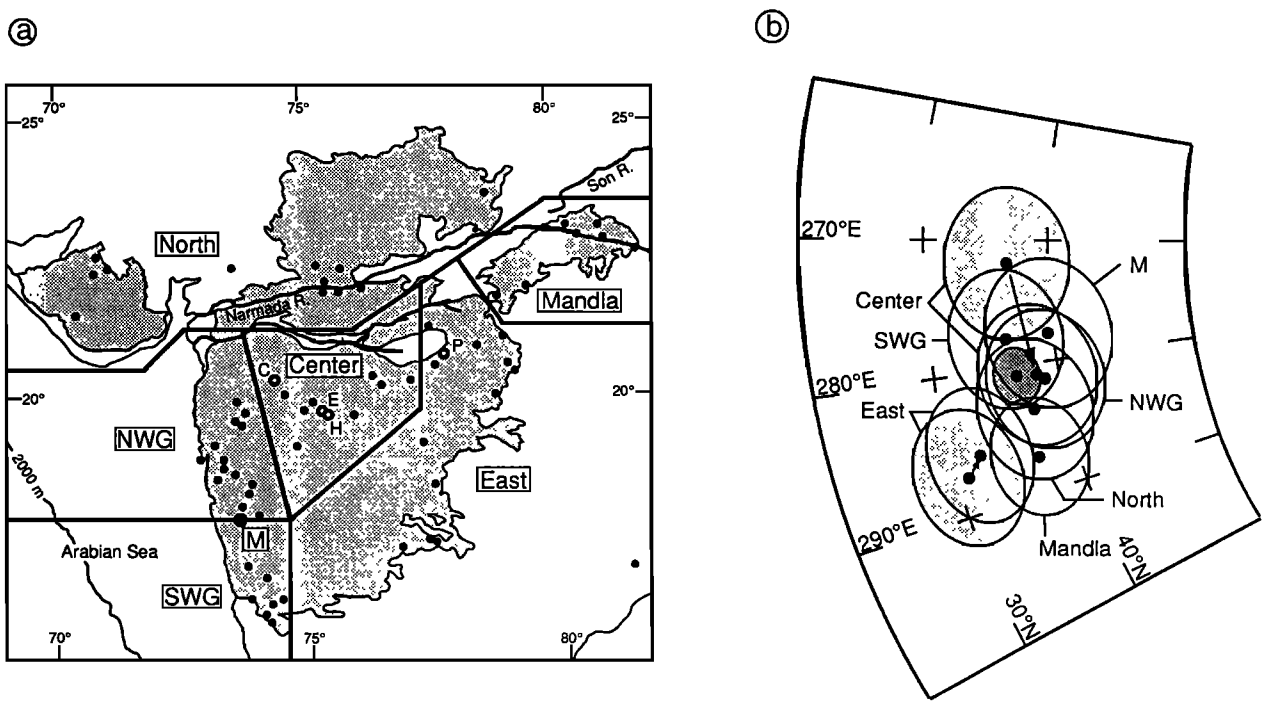


Figure 13. Separation of the sites in distinct geographical provinces for regional study. (*a*) Map of sites sampled for paleomagnetism (excerpted from *Rao et al.* [1985] updated) with ad hoc regional subdivision (C: Chandor; E: Ellora; H: Harsul; M: Mahabaleshwar; P: Pohor; NWG: North Western Ghats; SWG: South Western Ghats). (*b*) Equal area projection of the virtual geomagnetic poles averaged in each regional subdivision (Table 6) with their 95% confidence circles. The arrows indicate the changes of the center and east mean poles obtained after exclusion of Ellora, Harsul, and Pohor. The light shaded areas are the 95% confidence circles of the center and east mean poles prior to these correction, and the dark shaded area surrounds the overall mean pole (also prior to excluding Ellora, Harsul, and Pohor).

TABLE 7. Fisher and Bingham Statistics for Various Subsets of Deccan Paleomagnetic Poles Ellora, Harsul, and Pohor Sites Excluded

	Fisher						*Bingham*					
Subset	*N*	*Lat.*	*Long.*	*K*	A_{95}	*s.d.*	*Lat.*	*Long.*	*A13*	*A12*	*Ratio*	*Azim.*
					Mixed Polarity Flows							
All poles	485	35.7	281.3	20.3	1.5	18.0	35.8	281.5	1.4	1.5	1.13	125.6
$x_1 = 2, 3$ or 9	426	36.1	281.7	21.1	1.5	17.6	36.3	281.8	1.5	1.7	1.17	126.8
$x_1 = 2$ or 3	349	36.5	282.3	20.5	1.7	17.9	36.7	282.5	1.6	1.8	1.13	125.5
$x_1 = 3$	163	36.9	281.3	21.6	2.4	17.4	37.1	281.4	2.2	2.6	1.16	114.2
					Lower Normal Flows							
	11	34.3	284.1	9.9	15.3	25.9	34.2	285.6	12.8	13.9	1.08	140.3
					Reverse Flows							
All poles	361	34.2	280.8	20.8	1.7	17.8	34.2	281.0	1.5	1.8	1.17	116.2
$x_1 = 2, 3$, or 9	310	34.4	281.0	21.6	1.8	17.5	34.5	281.2	1.6	1.8	1.13	113.0
$x_1 = 2$ or 3	249	35.2	281.9	20.5	2.0	17.9	35.3	282.1	1.9	2.1	1.14	118.1
$x_1 = 3$	120	36.0	281.0	21.6	2.8	17.5	36.1	281.2	2.5	3.0	1.20	102.5
					Upper Normal Flows							
All poles	110	40.5	282.7	22.2	2.9	17.2	40.9	282.7	2.6	3.0	1.16	144.1
$x_1 = 2, 3$ or 9	104	40.7	283.6	23.6	2.9	16.7	41.1	283.6	2.5	3.1	1.24	144.0
$x_1 = 2$ or 3	89	40.0	283.3	22.8	3.2	17.0	40.4	283.3	2.8	3.4	1.24	145.3
$x_1 = 3$	39	39.5	282.8	22.2	5.0	17.2	39.9	282.0	3.7	5.6	1.50	137.5

The coefficient x_1: quality based on knowledge and value of confidence interval for the flow mean directions (see Appendix). Lower normal flows are labeled *N*2 in Appendix. Reverse flows are labeled *R*1 in Appendix. Upper normal flows are labeled *N*1 in Appendix. *N*: number of virtual geomagnetic poles (VGPs); Lat. Long., latitude and longitude of mean VGPs; *K*: Fisherian precision parameter; A_{95}; semiangle of the cone of 95% confidence; s.d.: angular standard deviation; A13, A12: semiaxes of the Bingham ellipse of 95% confidence; Ratio: elongation of the Bingham ellipse of 95% confidence (A12/A13); and Azim.: azimuth of the great axis of the Bingham ellipse of 95% confidence.

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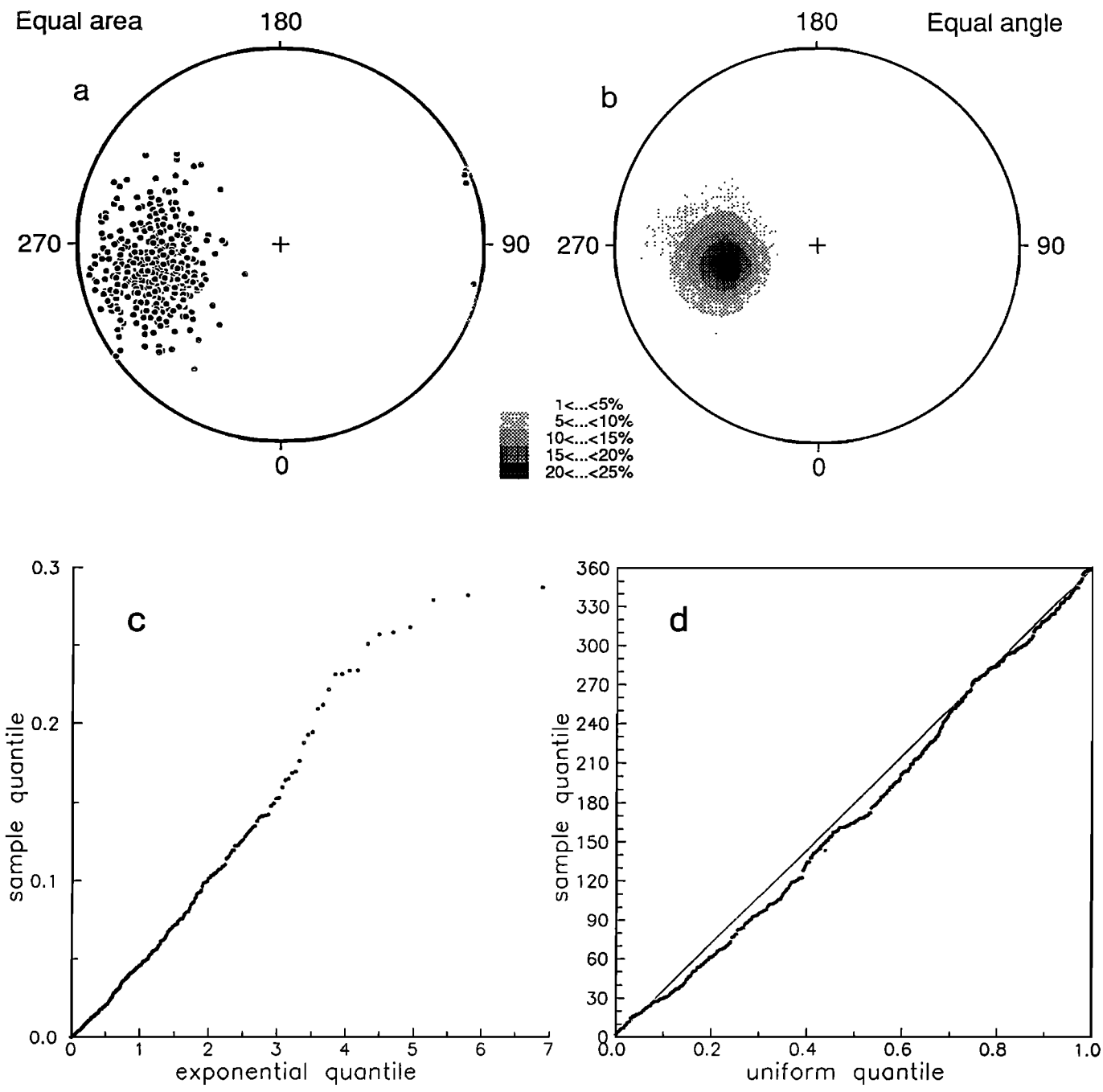
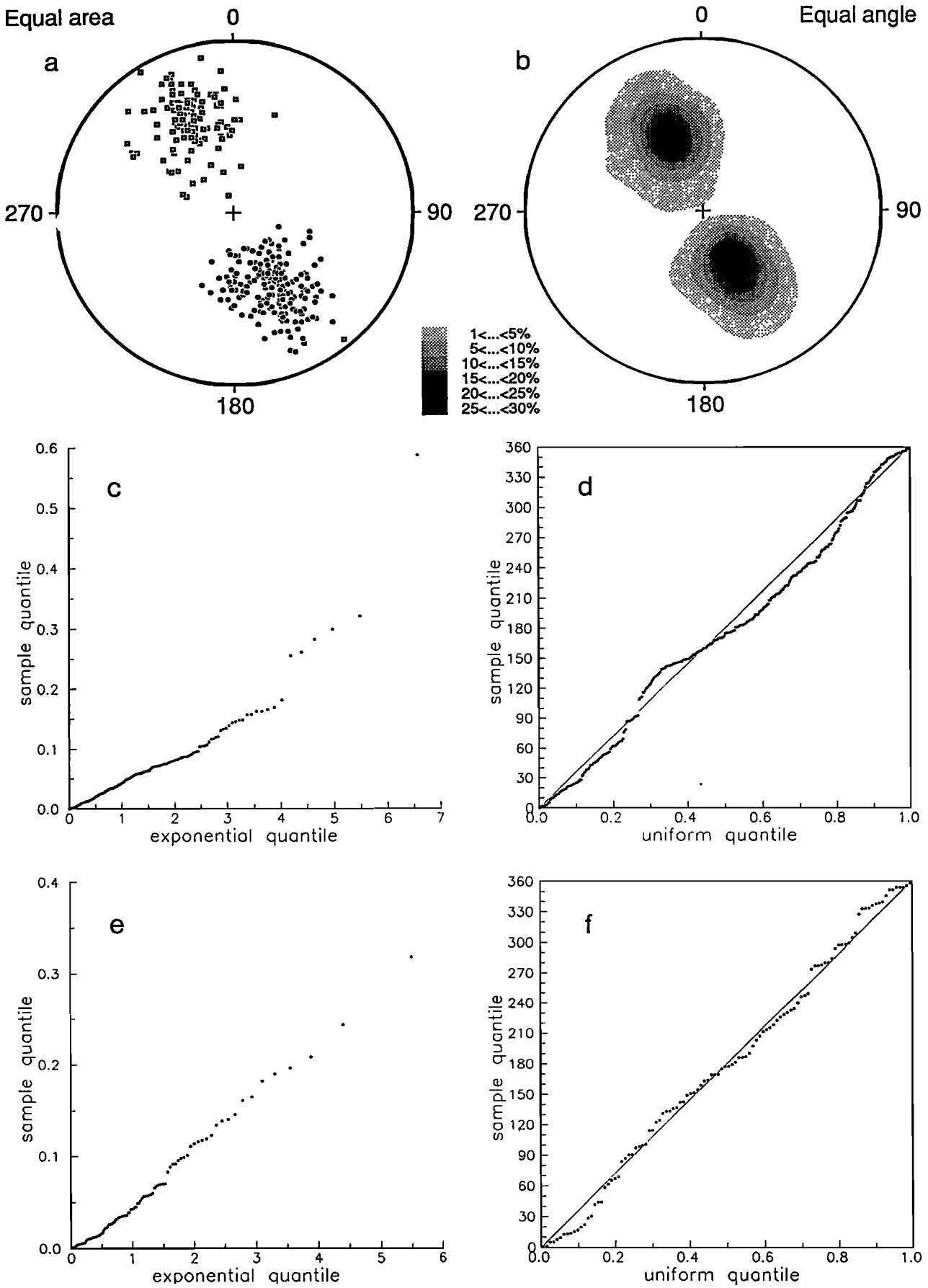


Figure 14. Same as Figure 11 when the sites of Ellora, Harsul, and Pohor (Table 7) and the transitional data are excluded. (*a*) Equal area projection. (*b*) Isodensity diagram. (*c*) and (*d*) Colatitude and longitude plots, respectively [*Fisher et al.*, 1987].

dence angles). The Pohor site has only normally magnetized flows. If we compare its altitude with those of Ellichpur on one hand and those of the Nagpur area on the other hand, it seems quite possible that the normal-reverse (*N-R*) transition was recorded here. We have therefore recalculated the regional statistics of the "east" and "center" provinces without these three sites. This yields a significant improvement of the regional VGP grouping (Figure 13*b*). When the overall statistics are redetermined, excluding these three sites (Table 7), the mean VGPs are

Figure 15. (opposite) Same as Figure 12, but with original polarities preserved. (*a*) Equal area (open squares in the upper hemisphere and full dots in the lower hemisphere). (*b*) Isodensity diagram (upper and lower hemispheres are superimposed). (*c*) and (*d*) "Colatitude" and "longitude" plots, respectively [*Fisher et al.*, 1987] of reversed directions. (*e*) and (*f*) "Colatitude" and "longitude" plots, respectively [*Fisher et al.*, 1987] of normal directions.

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shifted from 1° to 3° eastward and 0.2° northward, the standard deviations decrease by 1° or 2° without any significant change in confidence angles, the elongations of the Binghamian confidence ellipses become quite smaller and their azimuths rotate to about 120°. The colatitude and longitude plots (Figures 14) indicate a distribution that is now closer to being Fisherian (compare Figure 14*c* and 14*d* to Figure 11*e* and 11*f*). Some longitudinal elongation of the data cloud is still seen, particularly in the low isodensity contours (Figure 14*b*).

4.2.4. Influence of magnetic polarity. When the data (Pohor, Ellora, and Harsul sites being excluded) are broken down into separate normal and reversed sets, it is interesting that regardless of the quality index of the data subset being analysed, the number of reversed flows is always found to be around 75%. Following earlier suggestions, *Courtillot et al.* [1986*a*, *b*] have shown that over 53 magnetostratigraphic analyses of the Deccan lavas, amounting to a total of more than 11 km of sections, are consistent with the hypothesis that the traps were erupted in a rather short time sequence consisting of an upper normal (*N*1), a reversed (*R*1) and a lower normal (*N*2) chron. *Wensink* [1973] suggested the possibility of a lower reversed chron, at the bottom of the sequence, in the East of the Deccan traps. We find no conclusive evidence for this fourth chron based on a new magnetostructural analysis. [*Vandamme and Courtillot*, 1990; D. Vandamme et al., manuscript in preparation, 1991]. On the basis of altitude and local stratigraphy (see also D. Vandamme et al., manuscript in preparation, 1990) we have attempted to determine to which normal chron the sampled flows belong. Statistics for the various subsets of data are given in Table 7 and illustrated in Figure 15.

For the reversed data, which are the most numerous, the average directions provided by Fisher and Bingham statistics never differ by more than 0.2°, regardless of data quality. The Bingham ellipses have a small ellipticity, always smaller than 1.2 although the azimuth of the elongation is very consistent between 100° and 120°. The upper normal flows, which provide three times less data, show similar results, with maximum direction differences of 0.7°, elongation of 1.2 and azimuth between 130° and 145°. Only for the highest quality results does elongation reach 1.5 but there are only 39 data. In the case of the lower normal flows, there are far fewer results (11 data or 2% of the total) but these are completely consistent with the previous ones. Elongation is again seen to be more pronounced on the lower density contours and to vary in amount and direction with the density value (Figure 15*b*). The fit of the data to a Fisher distribution is illustrated by the test plots (Figure 15*c* and 15*d* for R1 data, Figure 15*e* and 15*f* for N1 data). We see in Table 7 that the mean reversed and normal directions move closer to being antiparallel as the quality index increases, until they become statistically indistinguishable with high confidence (Figure 17). The estimates of the Fisher precision parameter *K* are similar (ranging between 20 and 24). The angular distance between the mean normal and reversed poles, with the best quality selection, is 3.5° ± 5.7°. Should this difference be considered as significant, there are various ways in which it can be interpreted.

For instance, one could assume that a small amount of recent normal overprint still remains in the characteristic reversed and normal directions which have been considered as primary. Indeed, we have found many cases in our study where a recent overprint occurs and is not easily removed (for example, site 27, Figure 7; site 06, Figure 9). *Subbarao et al.* [1989] also emphasized the effect of alteration on the NRM in the Narmada region. Because of the magnetic cleaning methods usually used during the sixties and early seventies and of the way in which the characteristic directions were extracted, it is reasonable to assume that many earlier published directions are still somewhat biased by an overprint. Notice that the effect of this overprint may be partly removed in the process of data selection, which leads to an increase in the latitude of reversed flow VGPs and a decrease in that of normal flow VGPs (Table 7), whereas the longitude remains relatively stable. Let ±*I* be the inclination of the primary remanence acquired near 30°S, and i the inclination of the remaining (unremoved) secondary overprint, the intensity of this overprint being a fraction ε of the primary component (Figure 16). Let θ be the paleocolatitude corresponding to the primary direction; because of the overprint, VGPs based on normal samples will be closer to the site than the

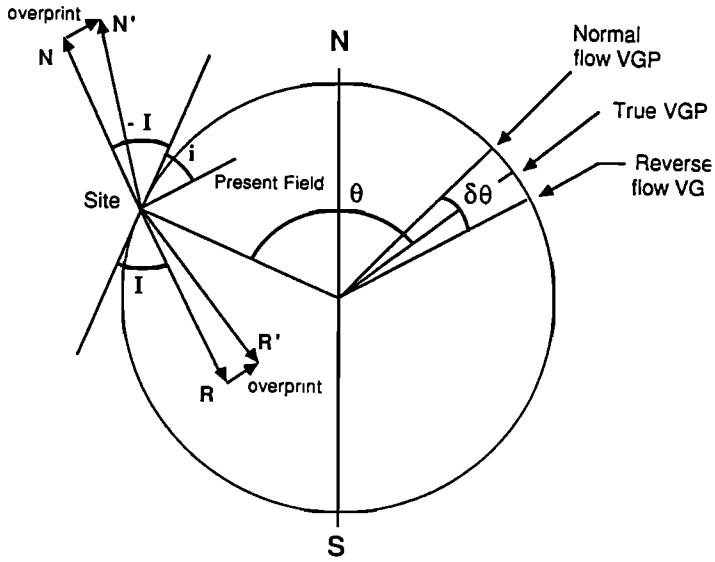


Figure 16. Schematic representation of the influence of a small recent normal polarity overprint on reversed (respectively normal) primary remanences acquired at a different latitude (see text). *R* and *N* are respectively the primary reversed and normal magnetic components, with inclination ± I. *R'* and *N'* are the components resulting from overprinting of *R* and *N* by a small component with inclination *i*. The paleocolatitude for the primary directions is labeled θ and *d*θ is the angular distance between the poles determined from the overprinted directions.

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"true" VGPs by an amount $\delta\theta/2$, while those based on reversed sample will be further away by the same amount. It is easy to show that, to first order in ε,

$$\delta\theta = \sin 2\theta\,(\sin i/\sin I + \cos i/\cos I)\varepsilon$$

Then,

$$\tan\theta = -2/\tan I \Leftrightarrow \sin 2\theta = -4\tan I\,(4 + \tan^2 I)^{-1}$$

With the appropriate inclination values $i \approx 40°$, $I \approx 50°$, we find $\delta\theta = 1.8\,\varepsilon$ (with $\delta\theta$ in radians). Therefore we can interpret the difference between the mean normal and reversed poles (for the best data with $\alpha 95 < 5°$) $\delta\theta = 3.5°$ = 0.06 rad as simply due to a remaining fraction $\varepsilon = 3.5\%$ of recent normal overprint. This fraction need only be 6% for the total data prior to any selection. These values do not appear to be unreasonable.

Another possible interpretation is that the latitude difference between the VGP from the upper normal chron and that from the main (older) reversed interval could be due to apparent polar wander. *Besse and Courtillot* [1991] have derived a synthetic apparent polar wander path for India (Figure 17), in which Indian data themselves have relatively little weight; the path is constrained by data from the Eurasian, North American, and African plates which are transferred across the Atlantic and Indian oceans (see also *Courtillot and Besse* [1987] and *Besse and Courtillot*, [1988]). In the period from 60 to 70 Ma, polar wander was about 1.7°/m.y. Hence the 3.5° difference might be due to a 2.1 m.y. time difference between the two polarity intervals.

Yet another possibility would be a slight difference in multipolar components of the averaged reversed geomagnetic field compared with the normal field.

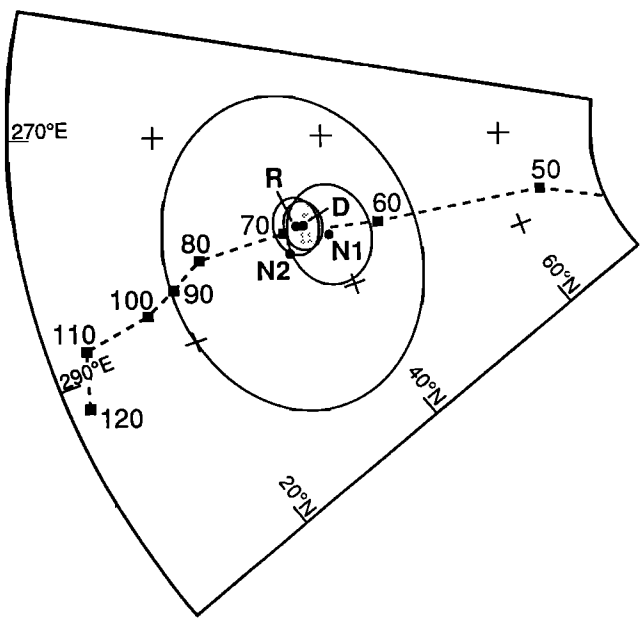


Figure 17. Comparison of the "best" paleomagnetic poles determined in the present compilation (Tables 7 and 8, data with $\alpha_{95} < 5°$) with the synthetic apparent polar wander path (APWP) for India determined by *Besse and Courtillot* [1991]. The full dots indicate virtual geomagnetic poles of flows from the main reversed sequence (*R*; 120 flows), from the upper and lower normal sequences (respectively *N*1; 39 flows and *N*2; 11 flows) and the overall mean (*D*; 163 flows) with their 95% confidence circles (shaded for the overall mean). The full squares show the APWP in 10 m.y. steps; the labels stand for the center of the 20 m.y. sliding window for each step.

4.3. Discussion

We wish to emphasize that, using the appropriate reversal test of *McFadden and McElhinny* [1990], the mean reversed and normal poles are not statistically different at the 95% probability level. Indeed, the angular difference between both poles is 3.5° and therefore smaller than the critical angle defined for this reversal test (γc = 5.7°). The classification as B in the terminology of McFadden and McElhinny is not meaningful for basaltic lavas because the recording of paleosecular variation (PSV) implies a relatively large critical angle. In any case the small angular difference is both in the right direction and has the correct order of magnitude to be interpreted in terms of a very small (3.5%) remaining normal overprint, or a short (about 2.1 m.y.) time difference between the main reversed and normal intervals, or a combination of the two effects. Table 8 summarizes the results from previous reviews. We note that the age subdivision into four groups proposed by *Pal and Bhimasankaram* [1971*c*] is based on a stratigraphic classification of intertrap frequency and is not consistent with the radiometric datings, paleontological information or magnetic reversal frequency reviewed earlier in this paper and must be considered obsolete. Three studies did emphasize the mainly *R-N* magnetostratigraphy of the Deccan traps [*McElhinny*, 1968; *Wensink and Klootwijk*, 1971; *Klootwijk*, 1979]. The first two papers included large individual studies (*Sahasrabudhe* [1963] and *Wensink and Klootwijk* [1971] respectively); the last is the most complete available review of paleomagnetic studies predating 1979. The estimate of the mean pole for the upper normal flows remains remarkably stable from 1968 to the present. The 95% confidence interval drops from 8° to 5°. In contrast, the estimate of the mean pole of the reversed flows evolves significantly: in particular, the pole latitude increases with publication date from 30° to 36° and the confidence interval drops below 3°. This evolution brings the mean normal and reversed directions into increasingly better antiparallelism. This general tendency of the reverse mean pole to come closer to (meaning antiparallel to) the normal one further strengthens our hypothesis of the presence of a remaining recent normal overprint, already suspected by *Klootwijk* [1979], and better accounted for through advances in demagnetization methods. We note for instance that samples from our site DK02 yield apparently acceptable results (K = 31.4, α_{95} = 8.3°) after "classical"

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TABLE 8. Comparison of Paleomagnetic Average Poles from Previous Reviews of Deccan Trap Data

Subset	*N*	*Lat.*	*Long.*	A_{95}	*dp*	*dm*
			McElhinny [1968]			
*N*1	29	37.	282.	8.		
*R*1	63	30.	281.	3.		
			Pal and Bhimasankaram [1971*c*]			
D	1	75.	271.	+2.8	2.	4.
C	14	57.	277.	+5.7	4.	8.
B	14	44.	285.	+5.3	4.	7.
A	54	30.	276.	+4.5	4.	5.
			Wensink and Klootwijk [1971]			
UWG	*5	39.1	282.7	+3.0	2.4	3.8
LWG	*5	32.7	275.1	+6.5	5.5	7.7
			Klootwijk [1979]			
*N*1	*32	38.9	281.4	5.4		
*R*1	*47	32.7	277.77	3.2		
*N*2	*6	29.6	298.5	18.5		
			Present Paper			
*N*1	39	39.5	282.8	5.0		
*R*1	120	36.0	281.0	2.8		
*N*1	11	34.3	284.1	15.3		

Subsets: *N*1: upper normal period. *R*1: main reverse period. *N*2: lower normal period. *A*: reversed period (assumed age about 70–75 Ma). *B*: mixed but mainly normal period (assumed age about 65 Ma). *C*: mixed but mainly normal period (assumed age about 60 Ma). *D*: tentative normal period (assumed age about 40 Ma). UWG: upper Western Ghats (mixed polarity). LWG: lower Western Ghats (reversed). *N*: number of flows (or sites if marked by an asterisk); Lat., Long., latitude and longitude of mean VGP; A_{95}: semiangle of the cone of 95% confidence (approximated as $(dp \times dm)^{1/2}$ for comparison where marked by a cross); and *dp*, *dm*: semiaxes of the oval of 95% confidence.

cleaning at 20 mT or 350°C, using Fisher statistics on characteristic components determined by least squares, whereas Halls' technique indicates that these directions are intermediate between the present field direction and the actual primary component, leading to relatively low estimates of declinations and high estimates of inclinations. The differences between normal and reversed flow poles previously found by *Verma and Mital* [1974] and *Wensink et al.* [1977] and interpreted as indicating respectively 23° and 14° of total northward drift of India during flow emplacement are clearly ruled out.

The foregoing analysis leads us to combine both reversed and normal directions in order to calculate a single overall mean pole. This final man pole for $N = 163$ flows with $\alpha_{95} < 5°$ (Table 7) lies at 36.9°N, 281.3°E ($A_{95} = 2.4°$). We note that the pole which we derived from only 16 flows (Table 6) is fully compatible with this, the angular difference between the two being $1.7° \pm 6.6°$. This agreement is partly due to chance, as is seen from the difference with our preliminary results [*Courtillot et al.*, 1986*b*]; the rather small angular standard deviation found at the time indicates that paleosecular variation had probably not been properly averaged out.

The "best" overall mean Deccan pole (together with the mean poles calculated separately for the upper normal, reversed and lower normal flows) is compared on Figure 1 to the synthetic Indian APWP of *Besse and Courtillot* [1991]. Although this pole was used in the calculation of the mean APWP poles, it has very little weight (1/13 for the 50–70 Ma time window, i.e., 60 Ma mean, 1/12 for the 60–80 Ma time window, i.e., 70 Ma mean and of course none for the others) in its construction. Yet the agreement is quite remarkable. The weighed ages of the data used in the 50–70 and 60–80 Ma windows being respectively 62.5 and 69.0 Ma, one can read an interpreted age of 67.2 ± 6.6 Ma (the age uncertainty being the quadratic sum of the uncertainties on mean poles on the Indian APWP and on our Deccan mean VGP), in excellent agreement with all age constraints summarized in section 3 of the present paper.

The foregoing discussion supercedes that of *Besse et al.* [1989], although it generally confirms their conclusions. A similar approach was taken by *Acton and Gordon* [1989], who transferred the previous pole of *Courtillot et al.* [1986*a*, *b*] to North American coordinates and compared this with a version of the North American APWP. As pointed above, the new overall mean pole found in this study is significantly displaced and has a significantly reduced confidence interval. Figure 18 shows where this pole would lie depending on its assigned age, as was done by *Acton and Gordon* [1989] in their Figure 2. The data are

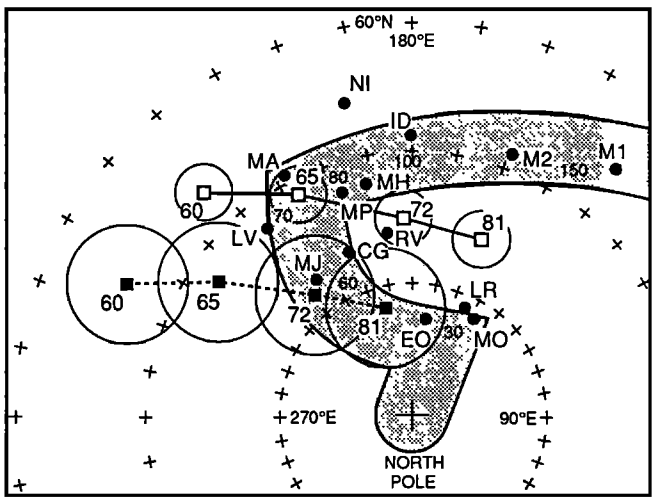


Figure 18. Overall Deccan virtual geomagnetic pole (VGP) from this paper transferred into North American plate coordinates, for comparison with the North American apparent polar wander path (APWP) from *Acton and Gordon* [1989, Figure 2]. The dots indicate the North American VGPs labeled with the same abbreviations as in Acton and Gordon. The shaded area envelops their estimated APWP labelled with their estimated ages. The open squares indicate where the Deccan pole would be transferred, should its age (in Ma) be as indicated next to each square symbol. The full squares correspond to the preliminary transferred VGP based on the 15 flows of *Courtillot et al.* [1986] and used by *Acton and Gordon* [1989].

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brought in best agreement in the 65–70 Ma time frame, hence the conclusions of *Acton and Gordon* [1989] stand. However, we find the comparison shown in Figure 17 more relevant, since far more data are integrated in the synthetic APWP.

Can the standard deviation of VGPs be interpreted as stemming solely from paleosecular variation? *McFadden and McElhinny* [1984] have proposed one of the most recent models (model F) for paleosecular variation (i.e., the angular dispersion of the paleomagnetic field directions), in which the dipole and nondipole parts of the field are assumed to be linked. This model compares favorably to paleosecular variation data available over the last 110 Ma. Model F predicts a VGP angular standard deviation of 13.4 ± 0.8° in the latitude range 25°–30° for the period 45–110 Ma, (using the same cutoff, 45° away from the mean, as we have done). The value of angular standard deviation which we find for the Deccan VGPs is 17.4 ± 1.4° (2σ) (calculated using a bootstrap method). This value therefore appears to be significantly larger than that predicted by model F (Figure 19); however, because of the very large amount of Deccan data, this may actually be the best available estimate of PSV at 65 Ma for that latitude range. In any case there is no clear evidence that the dispersion of Deccan poles involves noise in addition to natural intraflow dispersion and short-term secular variation of the dipole and nondipole parts of the field. It is likely that some data retained in our final compilation still contain some recent overprint and yet have been assigned in a high-quality index. But we have seen that a recent normal overprint should result in a decrease of the ellipticity of the quasi-Fisherian distribution of VGPs and not an increase in the dispersion of a strictly circular distribution.

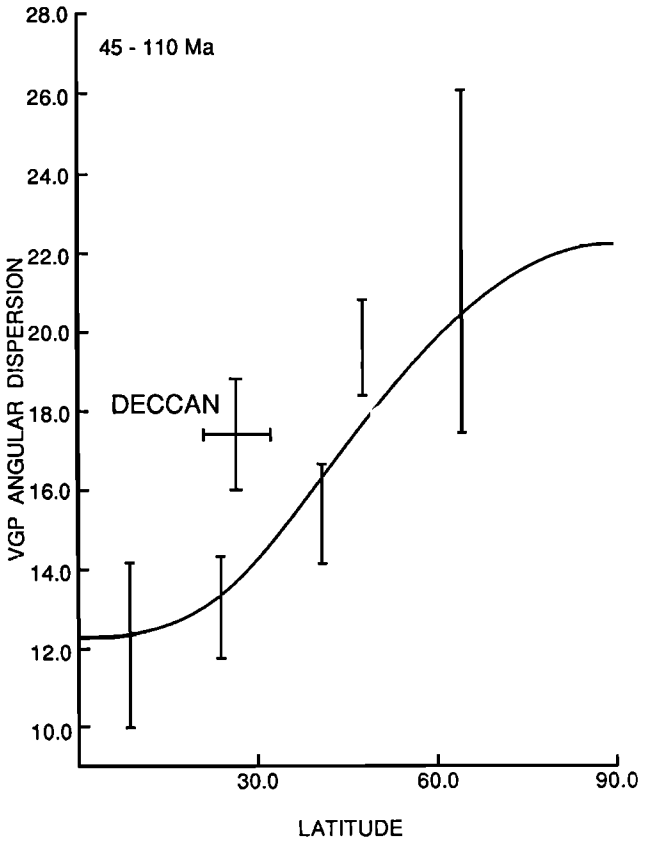


Figure 19. The estimate of angular dispersion of VGPs for the Deccan is plotted against angular dispersion expected from paleosecular variation as a function of paleolatitude (from *McFadden and McElhinny* [1984], Figure 9, appropriate for the 45–110 Ma time window).

5. CONCLUSION

The Deccan traps are probably the pre-Quaternary geological formation which has been most densely sampled for paleomagnetic studies and since the longest time. We have been able to list 563 paleomagnetic site-mean directions, each one corresponding to a single flow. Because of the diversity of scientific teams, times and places of sampling and wide extent of the flows, it is likely that the same flow has been sampled repeatedly but there is no direct and simple way in which such stratigraphic correlation can be established. We have studied the statistical dispersion of the data as a function of various selection criteria. A study of ellipticity of the data distribution using Bingham statistics shows that such ellipticity is negligible in most cases. It becomes larger only for the smallest data sets, such as the new data which we discuss in section 4.1 of this paper or the subset of the best quality data from the lower normal polarity interval of the Deccan magnetostratigraphy. In the former case, with 20 flows, the ratio of the axes of the 95% confidence ellipse is 2.1, but this drops to 1.5 when 4 doubtful or outlier data are discarded. In the latter, with 39 data it is 1.5. When the data set is broken down into separate geographic provinces, three out of nine subsets appear to display significant ellipticity (ratio of ellipse axes larger than 1.3). This is found to be largely due to outlying data, interpreted as transitional directions.

Outlying (transitional) data are hopefully removed by suppressing data which are more than 45° away from the mean. This efficient and classical (if somewhat unsatisfactory) criterion results in an increase of the Fisher precision parameter K from 9.6 to 18.3 (while N drops only from 563 to 519). Other estimates are hardly affected. When averages are computed for subsets of the data with increasingly better quality (quality being defined by an index which first specifies the use of demagnetization techniques, then the size of the 95% confidence interval), the K value remains stable at 17–18 (respectively 20–21 when the outlier sites of Ellora, Harsul, and Pohor are removed). Elongation of the ellipse is 1.2–1.3 (respectively 1.1–1.2) at an azimuth of 90°–110° (respectively 115°–125°) when the quality index is increased, the total number of data dropping from 519 to 180 (respectively 485 to 163). It is interesting that the direction of elongation of the data is approximately perpendicular to that which would be expected from either

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apparent polar wander or presence of unremoved recent overprints. However, we note that should paleomagnetic directions have been distributed in a Fisherian way, the VGPs would be expected to be elongated precisely along the observed almost EW direction with a ratio of semiaxes of 1.5. Since the observed values are somewhat smaller, actually halfway between 1.5 (Fisherian directions) and 1.0 (Fisherian poles), we cannot determine which physical parameter better approaches circular symmetry. The elongation is small enough that Fisherian averages and confidence intervals can be used in place of their Bingham counterparts (Tables 6 and 7). The 95% confidence intervals of the various subsets range from 1.5 to 2.6; larger values are obtained with smaller data sets. Our preferred overall mean pole is that determined when (1) data from Ellora, Harsul, and Pohor are excluded, (2) data further than 45° away from the mean are excluded, (3) only data with flow-mean 95% confidence interval less than 5° are included. This results in 163 data with mixed polarity (Table 7) and a mean at 36.9°N, 281.3°E (A_{95} = 2.4°).

Should one prefer to interpret the remaining spread (despite of the fact that it is quasi-isotropic and not elongated in the correct direction) as due to apparent polar wander or a remaining recent overprint, the corresponding estimates would lead to a duration of volcanism of 2.1 ± 3.4 m.y., or to a fraction of 3.5 ± 5.7% of overprint. However, the data dispersion (standard angular deviation) of 17.4 ± 1.4° may well result solely from natural intraflow and interflow dispersion due to cooling history and paleosecular variation. This value is not far, though slightly and significantly larger, than that predicted by PSV model F of *McFadden and McElhinny* [1984] at 13.4 ± 0.8°. Although very minor remagnetization and/or local rotation are likely to still affect even the directions in the higher quality subset, one could argue that for lack of other analyses with similarly large amounts of data, this might be the best estimate of PSV at 30°S latitude at 65 Ma.

We have compared the overall mean pole to the synthetic APWP of *Besse and Courtillot* [1991] transferred to India (Figure 17). The agreement is quite remarkable and, given the uncertainties in both the Deccan direction and the APWP determination, one can directly read an age for the Deccan from this comparison: this is 67.2 ± 6.6 Ma. Agreement is also found following the method of *Acton and Gordon* [1989], whereby the Deccan mean pole is transferred, under variable assumed age, to North America and compared to the North American APWP. The age is found to range between 62 and 72 Ma (i.e., 67 ± 5 Ma). However, use of the synthetic path of Besse and Courtillot, which also includes significant amounts of Eurasian and African data, should be both more accurate and more relevant.

The Deccan absolute age which is indirectly determined in this way is fully consistent with both the K-Ar and the ^{40}Ar–^{39}Ar absolute ages which have been reviewed in section 3 of this paper. The spread of the better determined plateau ages (Figure 5) for instance is between 63 and 68 Ma (i.e., 65.5 ± 2.5 Ma). This is also consistent with the range of the best estimates of the absolute age of the KTB, between 63.5 and 66 (at most 67) Ma. Paleontological data [*Jaeger et al.*, 1989] and magnetostratigraphic observations [*Courtillot et al.*, 1986*b*; *Gallet et al.*, 1989] further constrain the age and duration of Deccan volcanism. This is found to have started during a normal chron with the *A. mayaroensis* zone of the Maestrichtian (uppermost Cretaceous), which is therefore identified as chron 30*N*. Most of the volcanism (70% to 75% in volume if the total percentage of *R*/*N* data is a good indication) took place within reversed chron 29*R*, which is known to include the KTB itself. Volcanism continued within normal chron 29*N* and was over prior to planktonic foraminiferal zone P2. If also represented by the respective occurrences of normal versus reversed samples, chrons 30*N* and 29*N* would represent respectively 2% and between 20 and 25% of the time (volume) of eruption. Since chron 29*R* lasted approximately 0.5 Ma, the total volcanism would span slightly less than 0.7 Ma and might have started only a few tens of thousands of years prior to the 30*N*/29*R* boundary. These estimates are of course very coarse due to the highly episodic nature of volcanism. Yet, this timing is fully consistent with the stepwise character of many chemical and physical anomalies found in KTB sections (such as iridium, stable isotopes, disappearance levels of fossils, shocked minerals and so on—for an overview, see *Courtillot* [1990] and *Courtillot et al.* [1991]) and with the hypothesis of a causal link between the Deccan volcanism and the KTB events.

Should the hypothesis that most of Deccan volcanism occurred over less than 1 m.y. be correct, this provides excellent testing material for various geochronological techniques and for assessing the extent and influence of alteration. Finally, comparison of geochemical and paleomagnetic stratigraphy should become a preferred way to map in detail and laterally correlate the Deccan traps, and determine their structure [*Beane et al.*, 1986; *Devey and Lightfoot*, 1986; D. Vandamme et al., manuscript in preparation, 1991).

APPENDIX: A COMPLETE LIST OF PALEOMAGNETIC RESULTS FROM THE DECCAN TRAPS

The complete data set used for this study is listed in a file which can be sent by the authors to anyone interested (printed table or ASCII file on a personal computer floppy disk). References (code names are listed in a second file), locations (both precise geographic coordinates, and rough locations with code names listed in a third file), and upper and lower altitudes of sampled flows are listed when available, together with average paleomagnetic directions, VGPs and relevant statistics. For each result we have appended a two-digit "reliability" index, which is used in grouping and discussing the whole data base. The first

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TABLE A1. Paleomagnetic Results From the Deccan Traps

	Sampling Site					Characteristic Direction				VGP				Quality		
				Altitude												
Ref.	Site	Lat.	Long.	Lower	Upper	Dec.	Inc.	K	α_{95}	Lat.	Long.	dp	dm	x_1	x_2	P
DX59	PA	22.5	71.6	342	739	351.0	−16.0	...	3.5	58.1	268.6	1.9	3.6	0	0	*N*1
RR63	RA	17.1	81.8	...	...	327.0	−45.0	12.9	8.0	35.9	298.7	6.4	10.1	0	0	*N*1
SA63	AA	16.1	74.1	630	630	35.0	−70.0	...	...	13.8	233.8	...	...	9	8	*X*
BP68	CH	20.3	74.3	768	771	146.0	69.0	...	...	11.4	274.6	...	...	9	1	*R*1
AT70	AM	21.7	81.8	701	701	146.0	46.0	...	4.3	30.9	297.2	3.5	5.5	3	1	*R*1
PB7b	SA	23.9	78.6	536	557	37.0	46.0	266.7	4.7	56.6	154.7	3.8	6.0	3	9	*T*
KX72	MA	17.9	73.6	212	212	164.8	40.8	158.0	5.3	46.2	273.9	3.9	6.4	2	1	*R*1
VX73	JD	23.2	80.3	476	476	338.0	−23.0	250.0	4.0	48.8	294.1	2.3	4.3	3	1	*N*2
CX86	BU	20.5	76.3	...	...	154.9	47.2	788.7	2.4	35.4	283.6	2.0	3.1	3	1	*R*1
VXpp	RA	17.0	81.6	280	280	306.3	−46.6	96.6	5.4	21.3	311.5	4.5	6.9	2	1	*N*1

(An excerpt to illustrate the complete file, see text.) Ref. is a code name for the original paper in which the data was published, the two last digits standing for the publication date. Lat. is the latitude. Long. is the longitude, and altitudes are in meters. Dec. is the declination, Inc. is the inclination, K is the Fisher's precision parameter, and α_{95} is the semiangle of the 95% cone of confidence. P is the magnetic polarity of the flow, $N1$ is the upper normal period, $R1$ is the main reversed period, $N2$ is the lower normal period, T is for flow directions deemed by original authors to be intermediate ($x_2 = 9$), and X is for flow directions more than 45° away from the mean ($x_2 = 8$).

index x_1 is 0 when data have been obtained without magnetic cleaning, $x_1 = 1$ when cleaning was used, but α_{95} is larger than 10°, $x_1 = 2$ when $\alpha_{95} < 10°$, and $x_1 = 3$ when $\alpha_{95} < 5°$ (α_{95} limit values are twice as large for flow means), and $x_1 = 9$ when α_{95} is not listed. The second index x_2 is 0 when only a mean of several flow directions obtained by original authors is listed, $x_2 = 1$ for an individual flow direction, $x_2 = 9$ for flow directions which are deemed by original authors to be intermediate, and $x_2 = 8$ for flow directions which are more than 45° away from the mean and which we interpret as being unrecognized-intermediate directions. See Table A1 for illustration purposes. Original papers are all listed in the references.

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